



Advancements of phonetics in the 21st century: Intonation [☆]

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ABSTRACT

Intonation, the linguistic use of voice pitch, is critical for acquisition, phonetic organisation at the phrasal level and subsequent speech processing. By indicating syntactic constituency and encoding information structure and conversational implicatures, intonation is not only crucial for communication, but also for understanding grammatical phenomena such as information structure and syntax. Nevertheless, existing research has not always done justice to the roles that intonation plays in grammar, acquisition, and communication. Here, we first present an overview of what intonation is (and what it is not), and briefly discuss the essential tenets of the autosegmental-metrical (AM) theory of intonational phonology, currently the most influential and widely used approach to intonation. We critically review the standard methods of conducting intonation research within AM and present newer methodologies developed in response to shortcomings of the standard methods. We conclude by reviewing potential changes to AM in light of the findings stemming from these newer methods, followed by suggestions for future research.

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1. Introduction

As the term intonation is often used to refer to a variety of phenomena, we begin with some definitions that also explain the scope of the present work. Intonation, as understood in this review, refers to the linguistic uses of voice pitch. Specifically, intonation refers to changes in pitch, interpreted at the utterance level, which serve to chunk utterances into smaller phrases and (in many, though not all, linguistic systems) to highlight parts of the message within those phrases. Intonation is also used to express grammatical functions, such as information structure (focus type and domain), information status (givenness), syntactic grouping, and sentence modality (interrogative, imperative, declarative, vocative etc.).

Most importantly perhaps, intonation is critical for communication, as it allows listeners to make inferences, i.e., to indirectly read information from how something is said; e.g., when a speaker says to a receptionist “I have an appointment at nine” with rising pitch on *nine*, their interlocutor is likely to treat the utterance as a statement about the speaker’s appoint-

ment time but also infer that they are indirectly asking whether the appointment is in the common ground. If, instead, the speaker is asked for their available appointments and replies “I have an appointment at nine” with falling-rising pitch on *nine*, the interlocutor would infer that the speaker is offering an appointment at that time and indirectly asking if that works for them as well. In the linguistic domain, intonation, which is post-lexical, is sometimes confused with lexical tone and lexical pitch accent, both of which determine lexical or grammatical meaning along with the consonants and vowels that make up the words. Although these uses of pitch are of interest and certainly interact with intonation at the utterance level, they are not the focus of this paper.

In the phonetic domain, intonation is often used as a synonym for fundamental frequency (F0). F0, which is the main acoustic property of the speech signal that is perceived as pitch, is the chief exponent of intonation, but as we show later on, it is not the only one (see [section 6.2.5](#)).¹ F0 in and of its own is not interpretable without information as to where in the metrical (or prosodic) structure meaningful modulations take

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¹ As F0 and pitch are in a quasi-linear relationship for the range of frequencies related to intonation (Glasberg & Moore 1990; Hermes & van Gestel 1991; Nolan, 2003; for a review, see Hirst & De Looze, 2021), we will be using pitch and F0 interchangeably, although we stress that, strictly speaking, they are distinct.

place. Only when these are considered together do we have intonation. For instance, in languages that have stress, an F0 fall on a word's stressed syllable may indicate that the word is accented or highlighted, guiding the listener's interpretation of referential meaning, whereas a similar fall at the end of a word is more likely to indicate that the word is phrase final. Information regarding the location of stresses and phrasal boundaries is encoded in the language's metrical structure, which must be considered together with the F0 modulations to understand intonation. Similarly, in languages that do not have stress, access to prosodic structure which encodes phrasal constituency is necessary.² From here on, we will use the term *metrical*, as it is the established term in the model we focus on, but with the understanding that metrical encompasses prosodic, and that the differences hinge mainly on whether the structure encodes relative strength (informally, stress).

It is important to note that F0 does not encode only the grammatical functions that comprise intonation. It can also reflect cognitive styles (for attributes of F0 in relation to autism, see Wehrle et al., 2022). Moreover, it is used for paralinguistic and indexical purposes, such as indicating excitement and performing gender (on paralanguage, see Ladd, 2008: chap. 1; on pitch and gender, see, Daly and Warren, 2002; Yuasa, 2010, *inter alia*). Such effects are reflected in pitch span and dynamism, and occasionally in specific intonational choices. Most importantly, however, whether speakers produce their utterance with “monotone” or dynamic intonation, and with narrow or wide pitch span, they still have to adhere to the rules of and the choices afforded by their language's intonation system.

Finally, the term intonation is sometimes used as a synonym for prosody. Prosody, however, is a broader term that refers to a number of interrelated phenomena, namely stress, rhythm, and phrasing. These phonological phenomena are reflected in metrical structure which interacts with intonation. Thus, the term “prosody” as used here refers “to abstract prosodic structure that indicates the phonological organisation of speech units into higher-level prosodic constituents and their relative prominence within these constituents” (Steffman et al., 2022: 3).

In short, intonation is part of a language's phonology and interacts with the metrical component. Intonation and metrical

structure together play an important role in expressing syntactic structure, information structure and information status, and thus they interact with syntax, semantics, and pragmatics. Further, because intonation is also used to indicate sentence modality and illocutionary force and can lead to inferences about the pragmatic intent of an utterance, it plays an important role in spoken communication. We know, for instance, that neurodivergent population groups, such as speakers with autism, tend to have difficulties using and interpreting intonation patterns (see Grice et al., 2023, for a review). Similar issues apply to L2 learners for whom interference from L1 intonation can lead to communicative difficulties (e.g., Ortega-Llebaria & Colantoni, 2014). Finally, it is now understood that intonation is important for first language acquisition (see Prieto & Esteve-Gibert, 2018): intonation and rhythm are the components of speech that foetuses are exposed to in utero and infants attend to during the first year of acquisition (see Gervain, 2018, for a review). Research suggests that even infants' crying patterns are influenced by the intonation of their ambient language (Mampe et al., 2009).

The above strongly suggests that intonation should be studied in depth and that a basic understanding of intonation is essential for other areas of research. However, studying intonation is challenging. Annotation and analysis require considerable training because intonation is complex and, just like other areas of linguistics, cannot be researched without training. This training should, at a minimum, involve both understanding the phonological theory behind empirical studies and learning how to appropriately visualise and “read” intonation in the speech signal, both aurally, and visually by inspecting pitch tracks.³

In the remainder of this paper, we first present a brief overview of intonational phonology (section 2), followed by the essential tenets of the autosegmental-metrical (AM) theory of intonational phonology (section 3) and associated empirical evidence (section 4). These are followed by a discussion of empirical results that challenge certain aspects of AM (section 5), and the new methodologies and approaches they have led to (section 6). Finally, section 7 presents a brief overview of the current state of the art, as it emerges from the latest research, and section 8 identifies research topics that remain challenging. We note that in the past few years several review papers on intonation have appeared in volumes that cover prosody (on AM, Grice and Baumann, 2009; Arvaniti and Fletcher, 2020; Arvaniti, 2022a, with a commentary by Grice, 2022a, and a response by Arvaniti, 2022b). The present paper differs from earlier reviews in that it is forward looking, presenting not only traditional approaches but also recent advancements to the extent that space limitations allow.

The scope of this paper, given the extensive amount of work over the last two decades, is necessarily limited to the phonology and phonetics of intonation and how these can be investigated (and integrated). It does not cover recent work on intonation acquisition, or intonation processing, such as work investigating the processing time course via eye-tracking,

² *Metrical* and *prosodic* are terms that have been used with multiple meanings in the literature, while the information encoded in metrical and prosodic structures and the formalisms used to do so vary by model. For instance, Liberman & Prince (1977) used metrical trees with binary branching to capture only strength relationships pertaining to stress in English, and a type of metrical grid that focused on rhythm. Bracketed grids that encode both metrical strength and constituent structure (potentially including phrasal boundaries) were first proposed by Halle & Vergnaud (1987). The model of Nespor & Vogel (2007), known as prosodic phonology, uses unary branching trees with each node being labelled both for its constituent level and its relative strength. Pierrehumbert & Beckman (1988) used trees similar to those of Nespor & Vogel (2007), which they also call prosodic trees, though theirs do not include relative strength information since they were used for Japanese for which, Pierrehumbert & Beckman (1988) argue, metrical strength is not relevant. A full discussion of the merits of these and subsequent prosodic and metrical models and associated formalisms is beyond the scope of this paper. The interested reader is referred to the works cited above and to Beckman (1986), which is an early comparison of trees and grids. For our purposes, it is sufficient to state that intonation cannot be considered or phonologically represented by focusing exclusively on F0, as the relevance of F0 modulations – whether they are deliberate or incidental – is determined by the location of stresses (if present), and of phrasal boundaries. Thus, an understanding of metrical strength relationships, if relevant for the language under investigation, and of phrasal boundaries, which are relevant for all languages, is necessary, and this can be achieved by linking the representation of intonation to metrical or prosodic trees, as appropriate.

³ A new method, using periograms, which incorporate periodic energy into the visuals (thicker lines signalling more sonorous, and thus vocalic, parts of the utterance), makes this endeavour somewhat easier (Albert et al., 2022; Cangemi & Albert, 2023; see also examples in Grice 2022a).

mouse tracking or pupillometry, or neurophysiological correlates via Event Related (brain) Potentials (ERPs).

2. Why do we need a phonology of intonation

Intonation, especially when confounded with F0, is often treated as the study of melodies which are seen as *gestalts* or *configurations*. For instance, in the British School, the most important and indispensable element of the tune is said to be the nucleus, which is described in terms such as rise-fall, fall, fall-rise, and the like. This could suggest that it is possible to study intonation as a series of holistic pitch movements that can span entire utterances. Evidence that this approach is not observationally adequate is compelling, however (e.g., Grice, Ladd & Arvaniti, 2000; Arvaniti & Ladd, 2009; see Arvaniti, 2022a, for an overview).

We illustrate this point with the two utterances in (1) and (2) below. They both involve a final rise and can be interpreted as instances of the same melody, as they could both be answers that indicate the speaker is providing not an absolute number but a rough estimate.

-
- (1) *How many students are there in the class?*
 _[I am not sure, it's a bit over a hundred, something like] A
hundred and nine?
- (2) *How old is her dad?*
 _[I don't know exactly, about] *Seventy?*
-

Despite the similarities, the actual F0 contours are not identical, as can be seen in Fig. 1: in the longer utterance (Fig. 1a), the rise is localised on the rhyme of the final word, *nine*; in the shorter utterance in Fig. 1b, the first syllable of *Seventy* is low and the rise is realised on the following two syllables. Such observations of differences in the exact realisation of tunes and the ways in which they are timed relative to the text abound (see e.g., Ladd, 2008; Arvaniti & Ladd, 2009; Arvaniti, 2022a). They can only be accounted for if tunes are treated as compositional, i.e., as being composed of primitives of some sort. That is to say, certain component parts are obligatory and coordinate with specific elements of the segmental string, whereas others are incidental.

Further evidence of compositionality comes from the fact that intonation structure involves both syntagmatic and paradigmatic relations; the former relates to the placement of intonational categories (tones) at privileged positions within a phrase, the latter to the choice of tone at a designated position. An example of the latter is that one can produce the sentences in (1) and (2) with falls so as to indicate certainty about the number: the fall is still concentrated on the last word, *nine*, in the longer utterance (Fig. 2a) and the first syllable of *seventy* in the shorter one (Fig. 2b). An example of a syntagmatic change has to do with the location of the fall in (1); if the sentence in (1) was used to correct someone who thought there were only nine students in the class, the speaker would place a rise-fall on *hundred* leaving *nine* with flat F0 (Fig. 2c).

In short, as Ladd (2008:3-4) points out, intonation has phonological organisation. Just like consonants and vowels are organised into syllables and words, intonation consists of primitives that are organised into larger phrasal structures.

This structural aspect is not espoused by all approaches to intonation, many of which focus on phonetic detail and do not deal with abstraction. Abstraction is to some extent evident in the British School (see Nolan, 2022, for a review) and the IPO model ('t Hart, Collier & Cohen, 1990), but mostly in AM (Pierrehumbert, 1980; Ladd, 2008). Abstractions allow these theories, AM in particular, to better capture significant generalisations, such as similarities across instantiations of tunes, as well as similarities in intonation patterns across varieties of the same language and cross-linguistically.

Most importantly, a phonological account of intonation implies that there are abstract categories mediating the link between phonetic substance and meaning. Such categories are neither directly expressed as distributions of F0 values nor directly related to meaning (Cangemi & Grice, 2016; Arvaniti, 2019; Arvaniti, Katsika & Hu, 2024). For instance, a phonological approach would not relate phonetic parameters directly to information structure, such as relating a particular pitch range to a particular focus type. This is no different from the way we normally conceptualise lexical items as consisting of a semantic concept and a phonological form that is phonetically expressed. This conceptualisation means that we do not expect semantic concepts to be directly expressed as formant values or to be studied as such; rather, we expect semantic concepts to be mediated by phonological representations. By the same token, intonation and its uses cannot be reduced to the study of pitch contours without reference to their phonological representations.

3. Autosegmental metrical phonology in a nutshell

Currently, the most widely adopted phonological framework for studying intonation is AM, which is based on Pierrehumbert (1980). Though this work focused on English, AM has been adopted for the description and analysis of dozens of languages with appropriate modification (see, e.g., Jun 2005a and 2014a, and Gussenhoven & Chen, 2020, section IV). Below we briefly cover AM's key concepts, phonetic implementation module, and approach to intonational pragmatics, and discuss its advantages relative to other approaches to intonation.

3.1. AM phonology

As the theory's name implies, to understand intonation as conceptualised in AM we need to consider two distinct phonological components: an autosegmental string, which represents the intonational melody, and the metrical structure, which represents phrasing and, for languages for which this is relevant, the relative strength of constituents. According to AM, the primitives of intonation – the components of the above-mentioned string – are low and high tones, represented as L and H, respectively.

In AM, tones are *autosegments*, i.e., discrete phonological elements that associate with metrical structure. This association is an essential component of the model, as tones are interpreted in relation to metrical structure. Specifically, tones associate either with metrical heads (often stressed syllables) or with constituent boundaries. Association with a stressed syllable is indicated by the use of the star * notation (e.g., H*);

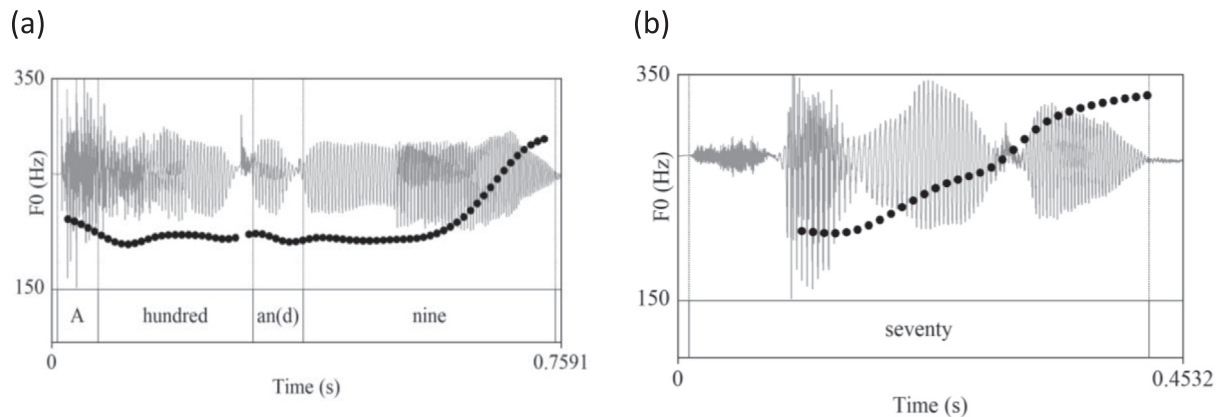


Fig. 1. Waveforms and pitch tracks of *A hundred and nine* (a) and *Seventy* (b), both produced with a rising tune by a native speaker of Standard Southern British English.

tones that associate with stressed syllables are known as *pitch accents*. The connection between stressed syllables and pitch accents is meant to imply that in AM, F0 is not treated as a phonetic correlate of stress but is determined by intonation, which is independent of stress. This is why Ladd (2008: 54) argues that pitch accents are *prominence cueing*, not *prominence lending*: they do not, in and of themselves, make words prominent; rather, their presence cues a word's prominence, which is derived from metrical structure (cf. Kochanski et al., 2005; Orrico et al., 2024).

In AM, pitch accents can be monotonal, e.g., L*, but may also consist of more than one tone, e.g., L+H*. In such bitonal and (occasionally) tritonal accents (e.g., Grice, 1995), one tone is typically starred. Starredness can be interpreted as indicating the strong node of a multi-tonal accent, i.e., the tone that is most resistant to coarticulatory pressures and likely to create the auditory impression the accent gives; e.g., American English L+H* is largely perceived as rising and its L tone can be undershot, while L*+H is perceived as low and rising and its L tone is *unlikely* to be undershot (Barnes et al., 2012). In practice, starredness is often treated as reflecting the temporal co-occurrence of the starred tone with the accented syllable (for a discussion on starredness, see Grice, 1995, Arvaniti, Ladd & Mennen, 2000, Gussenhoven, 2002).

Finally, in AM, the last pitch accent in the phrase is often referred to as the *nuclear accent* (and occasionally as the *focal accent*). The distinction between prenuclear and nuclear accents applies particularly to West Germanic languages, such as English and German, where the nuclear accent is also the only obligatory accent in a phrase (Pierrehumbert, 1994; Büring, 2016). In contrast, in some languages such as Greek and Spanish, prenuclear accents are not optional (Arvaniti & Baltazani, 2005, on Greek; Prieto, van Santen & Hirschberg, 1995, on Spanish), while in some Romance languages, such as Italian, the nuclear accent can be followed by postnuclear accents (also referred to as *postfocal* accents). AM does not formally distinguish between nuclear and other accents, in that in AM all accents can occur in all positions in a melody (Pierrehumbert, 1980; Silverman & Pierrehumbert, 1990). As this brief discussion shows, however, potential differences between prenuclear, nuclear and postnuclear accents and their respective distributions are language-specific and thus a matter of empirical investigation (see also section 3.3).

Tones that associate with phrasal boundaries are collectively known as *edge tones* and have a delimitative function, namely to cue constituent boundaries. Two or three types of edge tones are typically distinguished in a system, depending on the phrasal level with which the tones associate (thus, the number and nature of these levels is, to some extent, language-specific). For instance, English, German, and Greek are taken to have two phrasal levels, the intermediate phrase or ip and the intonational phrase or IP (Beckman & Pierrehumbert, 1986, Grice, Baumann & Benz Müller, 2005, and Arvaniti & Baltazani, 2005, respectively). In these systems, ips can contain one or more pitch accents, and IPs can contain one or more intermediate phrases. Another level of phrasing with tonal association is the accentual phrase, or AP, which typically includes one content word and its clitics. Languages analysed as involving an AP level include Japanese (Pierrehumbert & Beckman, 1988), Korean (Jun, 2005b), and French (Jun & Fougeron, 2000). These languages also have additional levels of phrasing, having been analysed as having edge tones at AP, ip and IP levels (Pierrehumbert & Beckman, 1988, on Japanese; Jun & Fougeron, 2002, and D'Imperio & Michelas, 2014, on French; Jun 2006, Jun & Cha, 2015, and Jun & Jiang, 2019, on Korean). The notation of edge tones varies, depending on the model; the most prevalent are Ta, T- and T%, for AP, ip and IP respectively, although subscript Greek letters are used for AP and IP as well (T α and T ι , respectively). In systems that have lexical stress and recognise edge tones associating with both the ip and the IP levels, these tones are known as *phrase accents* and *boundary tones* respectively. Phrase accents are considered a type of tone that may spread or show "stress-seeking" behaviour (see section 3.2). In systems in which edge tones associated with the ip-level do not spread but are locally realised, they are referred to as *boundary tones* (e.g., in descriptions of Korean, Bengali, and French intonation; Jun, 2005b, Khan, 2014, and D'Imperio & Michelas, 2014, respectively). Like pitch accents, edge tones may consist of more than one tone. Systems analysed as involving bitonal edge tones (e.g., HL%) include Japanese (Venditti, 2005), and Mexican Spanish (De la Mota et al., 2010). Analyses of Catalan (Prieto 2014), Japanese (Maekawa et al., 2002), and Bangladeshi Standard Bengali (Khan 2014) also involve tritonal edge tones (e.g., LHL%), while Jun (2005b) posits Korean IP boundary tones that

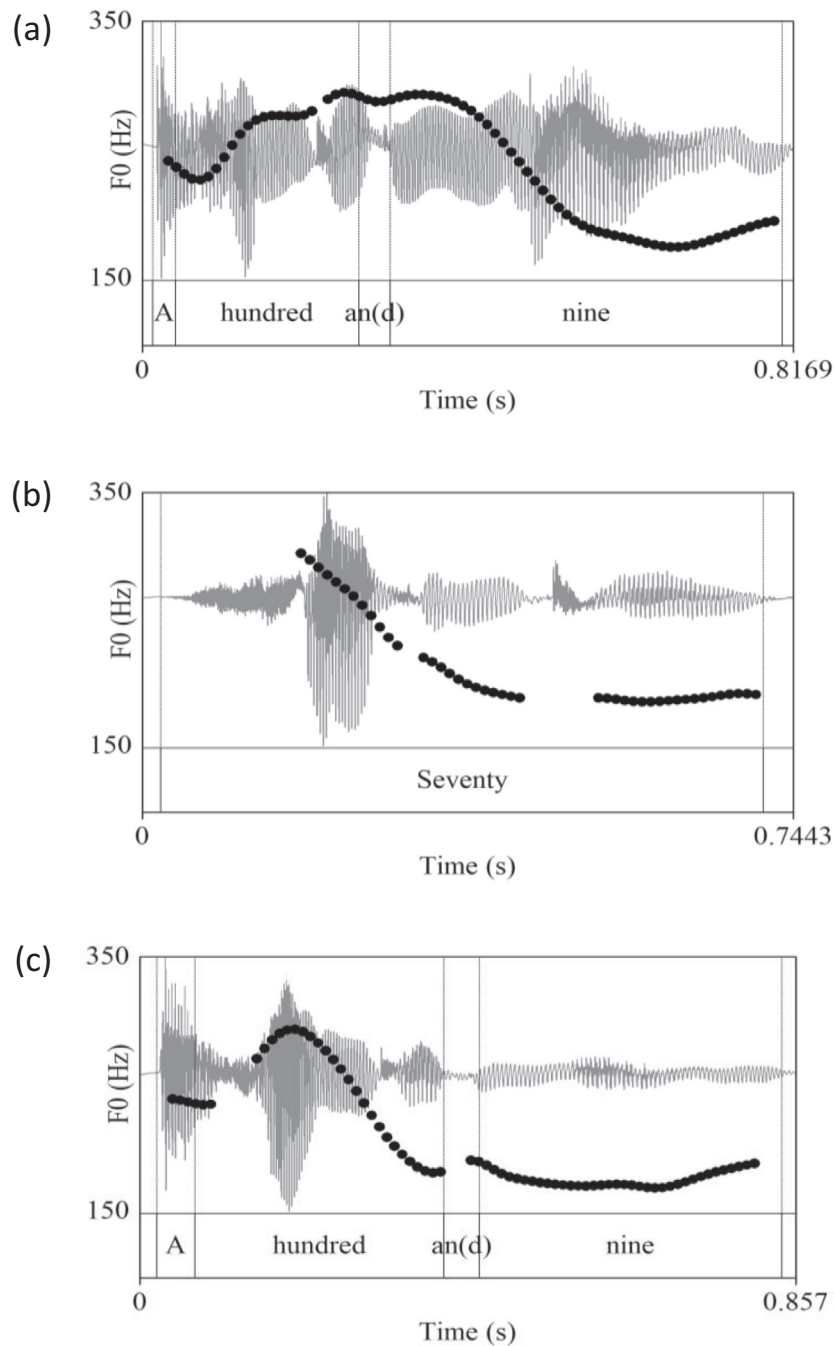


Fig. 2. Waveforms and pitch tracks of *A hundred and nine* (a), *Seventy* (b), and *A hundred and nine* with focus on *hundred* (c), all produced with a falling tune by a native speaker of Standard Southern British English. (The small final rises in (a) and (c) are due to exhalation and do not correspond to a rise in perceived pitch.)

consist of up to five tones, e.g., LH%, HL%, LHL%, HLHL%, and LHLHL%.

3.2. AM phonetics

AM is primarily a phonological theory, but it does address phonetic implementation. Indeed, the interaction between the phonological structure and phonetic implementation of intonation has played a critical role in the development of Laboratory Phonology (perhaps best illustrated in [Pierrehumbert & Beckman, 1988](#), which used empirical evidence from carefully controlled production studies to overturn earlier analyses of Japanese tone structure; see also Arvaniti, 2022: 79, for a

division of labour between phonetics and phonology in AM). The essentials of the standard AM understanding of intonational phonetics are briefly presented below.

The AM approach to how pitch contours are realised is sometimes referred to as *target-and-interpolation*, and the local minima and maxima that reflect abstract tones are treated as *tonal targets*. Abstract melodies are realised as sequences of tonal targets that are high or low in pitch relative to the context and reflect H or L tones, respectively. Targets are characterised by their *scaling* (F0 value, either absolute or relative) and *alignment* (temporal coordination with specific segmental events). Pitch accents align with the stressed syllable of the word they associate with, while edge tones align with the

edges of phrases and realised on sonorant segments adjacent to a relevant boundary (although some edge tones can align some distance from the boundary, see Gussenhoven, 2000, and Grice et al., 2000; see Fig. 2c for an example). The course of F0 beyond these targets is derived by interpolation. Interpolation is largely assumed to be automatic and monotonic, though other types of interpolation have also been proposed: for instance, Pierrehumbert (1981) proposed parabolic interpolation for some tonal sequences, including the much discussed but little investigated “sagging transition” between H* pitch accents in American English (for empirical evidence on the sagging transition in particular, see Ladd & Schepman, 2003). In fact, the sequence H*+H H* (rather than the later established H* H*) was originally proposed by Pierrehumbert (1980) for the representation of flat stretches in which a sagging transition between two high targets was absent.⁴

We use the tunes in Fig. 1(a-b) to illustrate the relationship between phonetics and phonology in AM. As shown in examples (3)–(4), the tune depicted in Fig. 1 is phonologically represented as consisting of a L* pitch accent followed by a H- phrase accent and a H% boundary tone. This applies whether the utterance consists of several words or just one. In both (3) and (4), since the utterance has broad focus, the L* pitch accent is phonologically associated with the stressed syllable of the last word, as dictated by metrical structure (Lieberman & Prince, 1977). The phrase accent H- and boundary tone H% are associated with the right boundary of the ip and IP, respectively (which are adjacent to each other).

For (3), the phonological association predicts that both the L* accent and the H-H% edge tones will be phonetically aligned with the stressed (and only) syllable of *nine*; in contrast, in *seventy*, L* aligns with the first (and stressed) syllable of that word, while the edge tones align with the last syllable. Consequently, in *nine*, the reflex of L* is a quick dip in F0 and the rise due to the edge tones is realised on that word’s rhyme. In *Seventy*, on the other hand, L* is associated with the first syllable of the word and the rise is realised on the last two syllables (cf. Figs. 1a and 1b). In short, in *A hundred and nine*, the entire tune is realised on one syllable, that of *nine*, even though there is plenty of preceding segmental material; the contour does not stretch to fit the utterance but, instead, is concentrated on the last word. These differences in realisation demonstrate the independence of the tunes from the segments (or *text*), and indicate that the realisation of a tune does not depend on the overall length of the utterance but only on those elements that are relevant for tune-text association; in our example, this element is the metrically strongest word which, by virtue of its metrical strength, bears the nuclear accent.

(3) [[a hundred and nine]ip]IP

| | |
L* H- H%

(4) [[seventy]ip]IP

| | |
L* H- H%

A similar situation obtains with the utterances in Fig. 2(a-b). Here, the tune is H* L-L% and the tune-text association is the

same as that shown in (3) and (4), only with different tones, as illustrated in (5) and (6). The tune in Fig. 2c, on the other hand, has a different structure in that the only pitch accent of the utterance, a L+H*, is associated with the word *hundred*, as shown in (7); thus, it is this word that shows a rise in pitch, while the L- phrase accent following it spreads, stretching over the rest of the segmental material.

(5) [[a hundred and nine]ip]IP

| | |
H* L- L%

(6) [[seventy]ip]IP

| | |
H* L- L%

(7) [[a hundred and nine]ip]IP

| | |
L+H* L- L%

The expectation within AM phonetics is that temporal and melodic coordinates of tonal targets, i.e., their alignment and scaling respectively, are relatively consistent for each tone type. The segmental landmarks with which the tonal targets of pitch accents are phonetically aligned are in or near the stressed syllable to which the accent is phonologically associated (Kohler, 1987, on German; Silverman & Pierrehumbert, 1990, for American English; Arvaniti & Ladd, 1995, and Arvaniti, Ladd & Mennen, 1998, 2000, for Greek; Ladd et al. 1999, 2000, on British English; Prieto et al. 1995, on Spanish; D’Imperio, 1995, 2000, 2001, on Neapolitan Italian).

These effects, illustrated in Figs. 1 and 2, do not apply only to utterances of different length; they also pertain to differences among phonological categories, particularly pitch accents. For instance, in the utterance “Only a millionaire!” in Fig. 3, the H tone of a L+H* accent is aligned with the stressed (and nuclear pitch accented) vowel of the word *millionaire* (which in American English is the first syllable); on the other hand, the same utterance produced with a L*+H pitch accent shows that it is the L tone that is temporally aligned with that same syllable. This changes the meaning from one of correction (L+H*) to one of incredulity (L*+H); for a discussion see Pierrehumbert & Steele (1989).

Similarly to pitch accents, the expectation for most edge tones is that they will co-occur with the last tone bearing unit (TBU) of the phrase, i.e., the phrase-final vowel (or potentially sonorant coda consonant, depending on the language). This appears to apply to all languages investigated so far, whether their intonation involves only edge tones or a combination of edge tones and pitch accents. A special case, attested in languages which have pitch accents, involves phrase accents. As mentioned earlier, unlike other melodic elements which tend to be localised, phrase accents may spread between the nuclear pitch accent and boundary tone (Lickley, Schepman & Ladd, 2005, on Dutch; Barnes et al., 2010, on American English). Spreading is illustrated in Fig. 4 with two Greek wh-questions, in which the stretch of low F0 – the reflex of a L-phrase accent according to Arvaniti & Ladd (2009) – between the utterance-initial L*+H accent and the final H% varies, depending on the segmental material intervening between these two tonal events (cf. the duration of the low F0 stretch

⁴ In Pierrehumbert (1980), the sequence was notated H*+H- H*.

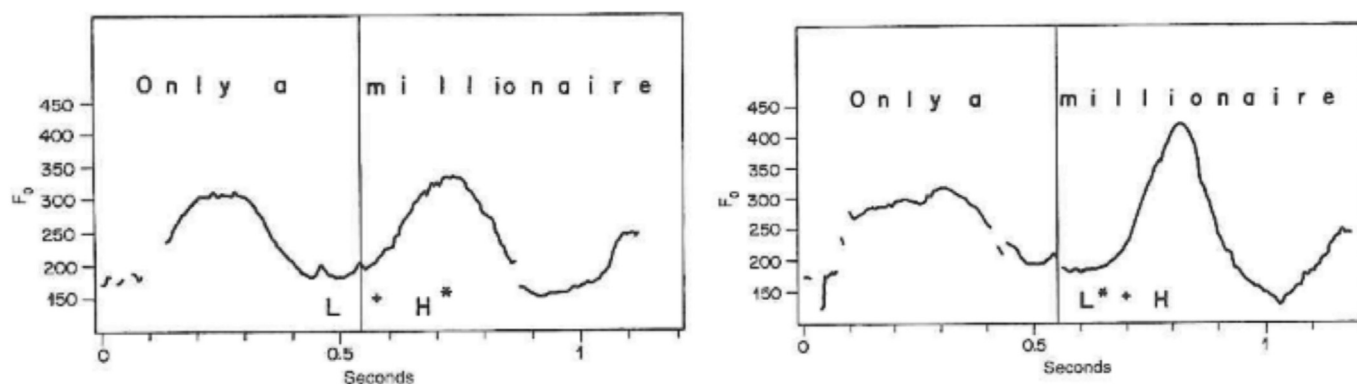


Fig. 3. Pitch contours of *Only a millionaire!* showing the difference in the timing of the L and H targets of L+H* (left) and L*+H (right); from Pierrehumbert & Steele (1989).

in Figure 2(a), where the last accent is on *nine*, and Figure 2 (c), where the last accent is on *hundred*). Additionally, phrase accents may show stress-seeking behaviour, a phenomenon analysed as *secondary association*. The mechanism of secondary association was first proposed by Pierrehumbert & Beckman (1988) for Japanese; it is discussed in relation to phrase accents in Grice et al. (2000) with English, Greek, Cypriot, Hungarian and Romanian data (see also section 6.1.2). Briefly, what Grice et al. (2000) have shown is that under certain conditions the phrase accents of specific tunes in some languages align with a stressed syllable, thus giving the appearance of being pitch accents. This is illustrated in Fig. 5 with examples from Greek: in Fig. 5a, the H- phrase accent of the typical polar question tune appears at a phrasal boundary, but when certain conditions are met, it occurs on the stressed syllable of the final word (Fig. 5b). We note that despite the appearance of such melodic elements that mimic pitch accents, words that carry them are not considered accented or prominent by native speakers (Arvaniti et al., 2006a, Ladd and Arvaniti, 2023, Orrico et al., 2024).

3.3. The treatment of intonational meaning in AM

Intonation meaning is largely seen as related to pragmatics. As Hirschberg (2006: 515) remarks, “since the interpretation of such intonational variations [accentuation and phrasing] is indeed dependent upon contextual factors, we will define intonational ‘meaning’ as essentially pragmatic in nature”. Hirschberg goes on to discuss several ways in which intonation influences the interpretation of syntactic and discourse phenomena (such as syntactic structure, and speech acts, respectively). Hirschberg’s account focuses on languages like English that use intonation both for phrasing and accentuation purposes, but similar pragmatic uses of intonation are reported in languages that do not use accentuation, such as Korean (e.g., Park, 2012). Based on our current understanding, accounts like Hirschberg (2006) and Pierrehumbert & Hirschberg (1990) discussed immediately below can be treated as maximal models of the pragmatic role of intonation cross-linguistically.

Perhaps the best-known model of intonational meaning within AM is that of Pierrehumbert & Hirschberg (1990) who provided a compositional analysis of intonation meaning for the categories that Beckman and Pierrehumbert (1986)

posited for American English (see also Büring, 2016, and Steedman, 2000). The general principles should be applicable to intonational systems at large, though so far, they have not been put to the test to a sufficient degree, with many studies falling into the category that Westera, Goodhue & Gussenhoven (2020: 441) call “specialist theories”, i.e., theories that “aim to account, in considerable detail and often with formal explicitness, for the usage of a narrow range of contours”. These include, among others, Bartels (1999) on English statements and questions, Savino & Grice (2007; 2011) on Bari Italian questions, Prieto & Borràs-Comes (2018) on Catalan questions, and Arvaniti, Gryllia & Baltazani (to appear), on Greek non-canonical wh-questions.

A point of discussion relates to whether such compositional accounts should attribute meaning to each melodic element. This applies particularly to pitch accents: for instance, in contrast to Pierrehumbert & Hirschberg (1990), who assume that all components of English tunes are meaningful (a view also espoused by the British school of intonation), Büring (2016) has argued that prenuclear accents are ornamental in English, while Calhoun (2010) posits that their presence is primarily motivated by metrical considerations. In some languages at least, the choice of prenuclear accent type appears to be restricted by utterance modality; this has been shown for German (Petrone & Niebuhr, 2014; Roessig, 2024) and Greek (Baltazani et al., 2015). Similarly, languages that do not have stress do not use pitch accents, and their intonation systems only mark phrasal boundaries. In such languages, pragmatic functions taken on by pitch accent type and placement in languages such as English may be assumed by phrasing and the choice of edge tones (e.g., Park, 2012, Jeon & Nolan, 2017, Hatcher et al., 2024, on Korean). The same may apply in some languages with lexical tonal phenomena, including languages with lexical pitch accent (e.g., Pierrehumbert & Beckman, 1988, Venditti, 2005, on Japanese) and languages with lexical tone (e.g., Chen, 2010, on Mandarin Chinese).

Since an account of intonational pragmatics is beyond the scope of this paper, we refer the interested reader to overviews in Prieto (2015) and Westera et al. (2020). What is clear is that the flexibility needed for interpreting intonational meaning, although implicitly accepted in many approaches to intonation, is only clearly and explicitly addressed in AM.

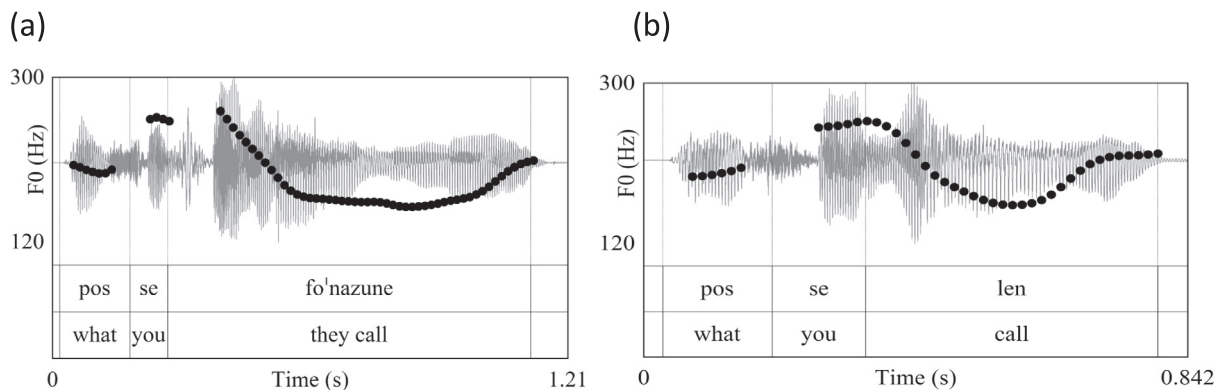


Fig. 4. Waveforms and pitch tracks of two Greek wh-questions (both of which can be loosely translated as “what’s your name?”), uttered by the first author using the same tune, autosegmentally represented as $L^*+H\ L^- H\%$, and illustrating the spread of the L^- phrase accent, which results in a longer low F0 stretch in (a), where the nuclear L^*+H pitch accent on [pos] is separated from the final syllable by more segmental material than in (b).

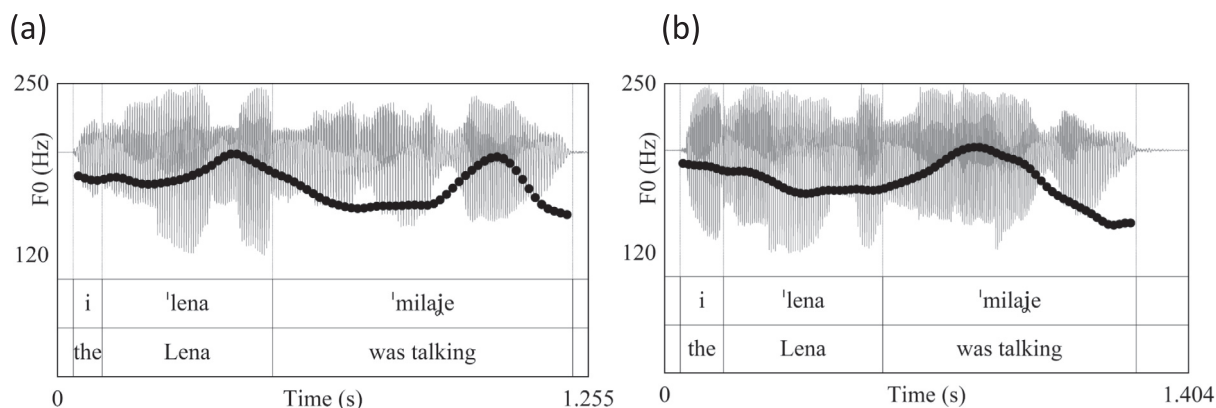


Fig. 5. Waveforms and pitch tracks of string-identical Greek polar questions, both produced by the first author with the same tune, autosegmentally represented as $L^* H^- L\%$, but with focus on the verb (/‘milaje/ “was talking”) in (a) and on the subject (/‘lena/ Lena) in (b); (b) illustrates the stress-seeking behaviour of the H^- phrase accent, which is realised on the stressed syllable of /‘milaje/; in contrast, in (a), the same H^- phrase accent is realised on the last syllable of /‘milaje/.

3.4. Advantages of AM

The AM conceptualisation of intonation as a string of abstract tones with a specific phonetic realisation offers several advantages. Consider, once more, the tunes in Fig. 1, *A hundred and nine* in 1a and *Seventy* in 1b. As noted, both involve a rise and both can be interpreted as instances of the same abstract melody, $L^* H-H\%$. By treating these contours as distinct realisations of the same tune, AM achieves the following:

- It provides a significant generalisation, namely that although these two tunes are superficially different, they underlyingly have the same representation, $L^* H-H\%$.
- It provides an explanation of the phonetic differences between the tunes, based on the independence of tune from text.
- The examples in Fig. 1(a-b) and their AM representations in (3)-(4) illustrate another fundamental point about AM, which is that neither its phonological representations nor its phonetic realisation component are meant to fully describe or interpret the F0 contour: in the longer utterance ((3) and Fig. 1a), several syllables have no phonological specification for intonation. In these cases, the F0 trajectory between tonal targets is derived by interpolation and thus, although F0 may show substantial excursions, these are not reflected in the phonological representation, do not involve tonal targets, and are not considered critical for the interpretation of the tune (for a full discussion, see Arvaniti & Ladd, 2009).

- An additional advantage of this conceptualisation of the relationship between tune, text, and metrical structure, is that it disentangles stress from intonation and from assertions such as that “stressed syllables have high pitch”. Although this applies in many instances (e.g., to *hundred* in Fig. 2c), it is perfectly possible for stressed syllables to have low pitch. In AM, this is not an issue, as the F0 level of a given syllable is not due to whether it is stressed or not, but to the tune, specifically the choice of pitch accent: stress, on the other hand, is determined by metrical structure which is independent of the tune. This is illustrated in Fig. 1, in which the stressed syllables of both *hundred* (Fig. 1a) and *seventy* (Fig. 1b) have low pitch (see Beckman & Edwards, 1994, and Ladd & Arvaniti, 2023, on the separation of stress from intonation).
- Further, the annotation system of AM provides flexibility in the representation of tonal elements; e.g., the use of bitonal and tri-tonal accents combined with the star notation allows for distinctions such as the difference illustrated in Fig. 3, between L^*+H and $L+H^*$ in American English. More generally, the compositionality of tunes makes it possible to account for the adaptability of the tune to the text, and this in turn allows AM to provide elegant and parsimonious analyses of complex F0 patterns such as those discussed in section 3.3 (see also section 4).
- Finally, the fact that in AM tunes are composed of distinct elements allows for a compositional approach to meaning as well, as touched upon in section 3.3. Such an approach to meaning allows for greater flexibility and better fits the observation that the relationship between tunes and meaning is many-to-many

(see Cangemi & Grice, 2016, for a discussion, and Arvaniti et al. (to appear), for a analysis of pragmatic meaning along these lines, using spontaneous speech).

The above points represent the main advantages of AM relative to other approaches to intonation mentioned above, such as the IPO and the British School, as well as other approaches not discussed in detail but alluded to, such as configurational approaches and approaches that rely on the full representation of the tune (for discussions, see Arvaniti & Ladd, 2009; Arvaniti, 2022a; Arvaniti et al., 2024).

4. Empirical evidence for AM

Empirical evidence in support of AM comes from both descriptive and experimental studies. First, descriptive studies show that, contrary to concerns sometimes expressed,⁵ the principles and building blocks of AM can be used to describe the intonational (and tonal) systems of typologically diverse languages. This is illustrated in many studies, but most particularly in Jun (2005a) and Jun (2014a), two volumes in which a total of 28 languages are prosodically described using AM principles (see also the overviews of intonation systems in Gussenhoven & Chen, 2020). AM descriptions include typical Western European languages that have stress (e.g., Dutch; Gussenhoven, 2004, Gussenhoven, 2005), languages with lexical pitch accent, such as Japanese (Venditti, 2005) and Basque (Elordieta & Hualde, 2014), lexical tone languages, such as Cantonese (Wong, Chan, & Beckman, 2005), languages that have neither stress nor lexical-level tonal phenomena, such as Korean (Jun 2005b), Malay (Maskikit-Essed & Gussenhoven 2016), Mongolian (Karlsson 2005), and Tashlhiyt Berber (Grice, Ridouane & Roettger, 2015), as well as descriptions of dialectal variation within linguistic systems (e.g., Fletcher, Grabe, & Warren, 2005, and Grice, German, & Warren, 2020, on English; Igarashi, 2014, on Japanese).

Experimental studies, particularly towards the end of the 20th century, focused on tonal alignment, leading to the Segmental Anchoring Hypothesis (henceforth, SAH) about the phonetic realisation of melodic elements. Following early work by Silverman & Pierrehumbert (1990), and Prieto, van Santen & Hirschberg (1995), Arvaniti, Ladd & Mennen (1998) found that Greek prenuclear accents involve an F0 dip at the onset of the accented syllable and a peak just after the onset of the first post-accentual vowel, as in the accent on [ko'rali] 'coral' in Fig. 6, which extends from [r] to the beginning of the postaccentual [i]. Arvaniti et al. proposed that the onset and offset of the rise are tonal targets that reflect underlying L and H tones, respectively, and form a L*+H accent. Subsequent research by Ladd and colleagues supported the idea of stable scaling and alignment of tonal targets in English and Dutch (Ladd et al., 1999, and Ladd, Mennen & Schepman, 2000, respectively), giving rise to the SAH which posits that the tonal targets of any melodic unit are timed with (i.e., 'anchored' to) specific locations in segmental structure, such as stressed

vowel onsets and offsets. Finally, the fact that Arvaniti et al. (1998) provided evidence for two distinct anchors, one for each tone, in the rising accents they investigated motivated the representation of rises as consisting of two level tones, L and H. In turn, this finding supported the AM view that the intonation primitives are levels, and not configurations such as rises and falls.

The idea of segmental anchoring stimulated research on tonal alignment in a variety of languages including languages with tone and lexical pitch accent (among many, D'Imperio, 2002, on Neapolitan Italian; Myers, 2003, on Kinyarwanda; Elordieta & Calleja, 2005, on Basque Spanish; Elordieta & Hualde, 2003, on Basque; Arvaniti & Garding, 2007, on American English; Dalton & Ni Chasaide, 2005, on Irish; Gordon, 2008, on Chickasaw; Prieto, 2009, on Catalan; Remijsen, 2013, on Dinka; Remijsen & Ayoker, 2014, on Shilluk). Based on this body of evidence, by the early 21st century the target-and-interpolation model was firmly established and the SAH widely accepted (D'Imperio, 2006, 2012 for reviews).

The importance of tonal alignment was first supported by the findings of imitation studies, in which stepwise changes in peak locations, as in the two examples of *millionaire* in Fig. 3 from Pierrehumbert & Steele (1989), lead to an abrupt shift in pitch accent category production (assertion vs. incredulity). The categorical effect of tonal alignment was perceptually investigated by D'Imperio and House (1997) and D'Imperio (2000) in Neapolitan Italian, and by Niebuhr (2007) in German, showing that tonal alignment perception effects can be independent from pitch height effects. However, other studies report a strong interrelationship between pitch height and peak timing: House (2002) demonstrated that raised F0 combined with a rightwards focal peak displacement shifts perception from statement to question intonation in Swedish (provided the focal accent is in final position). Similar results distinguishing statements from echo questions in Swedish were reported by House (2003).

Finally, research has also uncovered strong links between F0 and articulatory gestures, showing for example that accental F0 peaks may be timed to co-occur with articulatory gestures, such as minima and maxima of consonantal trajectories or peak velocity within the consonantal gesture of the syllable onset (D'Imperio et al., 2003, 2007; Niemann, Grice & Mücke, 2011, Mücke et al., 2012, Katsika et al., 2014; see Byrd & Krivokapic, 2021, for a review). The combined analysis of both kinematic and acoustic data led to the proposal that gestural tonal primitives for intonation are coordinated (*coupled*) with constrictions relating to metrically strong syllables or phrasal junctures, rather than being coordinated with acoustic landmarks. Although the evidence for the coupling of intonational targets and articulation is strong, it is somewhat limited and does not necessarily reduce the importance of establishing links between tonal events and acoustic landmarks. Even if the coordination is between articulatory gestures and tonal events, their relationship would likely be reflected in the acoustic signal as well. Thus, the two approaches are not mutually incompatible. Moreover, a coupling of articulatory kinematics with F0 should be treated with some caution, as the former represents the movement of the articulators with a fair degree of accuracy, while the latter is a noisy estimate of F0 derived from the acoustic signal by

⁵ Concerns are sometimes expressed that AM is not suitable for a given language because this language is prosodically different from English. Often, the origin of such concerns is a confusion between AM and ToBI (Tones and Break Indices), a system based on AM and devised originally for the prosodic annotation of American English (for a discussion of the differences, see Arvaniti, 2022a; for an overview of ToBI, see Jun 2022; for criticism of ToBI, see Ladd, 2022).

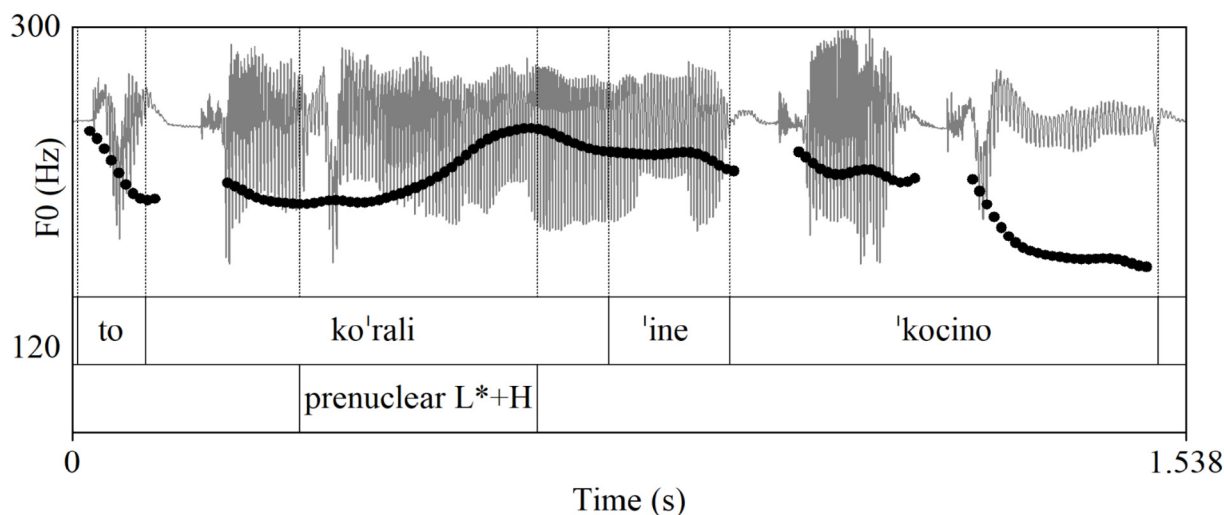


Fig. 6. An utterance ("the coral is red"), as produced by the first author, illustrating the alignment of the L*+H pre-nuclear accent of Greek, first described in Arvaniti et al. (1998).

algorithmic means (the same of course can be said about strict versions of the SAH, a point we return to in the next section).

5. Challenges to the standard AM model

5.1. Problems with standard methodologies: Empirical adequacy

As mentioned in section 4, a popular approach to the study of intonation within a (broadly construed) AM framework has been based on the SAH and the search for tonal targets that show stable alignment with segmental landmarks, on the assumption that invariable tonal targets reflect underlying tones. This approach has led to a wealth of new data on the intonation of scores of languages, but at the same time, it is becoming increasingly clear that a strong interpretation of the SAH cannot be supported.

In its strong interpretation, the SAH is taken to mean that the coordinates of tonal targets are invariant. However, empirical evidence has shown from the very start (e.g., Silverman & Pierrehumbert, 1990, van Santen & Hirschberg, 1994), that the realisation of tonal targets depends on a number of phonetic and phonological factors, including speaking rate (Fougeron & Jun, 1998, on French, Prieto and Torreira, 2007, on Spanish), syllable structure (D'Imperio, 2000, on Neapolitan Italian, Prieto and Torreira, 2007, on Spanish), phonological weight (Schepman, Lickley & Ladd, 2006, on Dutch), and the identity of following edge tones (Cangemi & Grice, 2016, on Neapolitan Italian). Segment-induced alignment differences have also been reported, e.g., for Pisa and Bari Italian (Gili Fivela & Savino, 2003), the French LH* rise (Welby & Loevenbruck, 2005), for Egyptian Arabic (Hellmuth, 2005), Neapolitan Italian (D'Imperio et al., 2007, Petrone & D'Imperio, 2015), and Tashlhiyt Berber (Grice, Ridouane & Roettger, 2015).

Perhaps the factor that has the strongest effect on tonal alignment stability is *tonal crowding*, the presence of more tones than TBUs, illustrated in Fig. 1b (example (3)). Tonal crowding can lead to several changes in tune realisation, including the following: compression, i.e., adjustments in the rate of F0 change (Arvaniti et al., 2017, on Polish); truncation

by means of target undershoot (Grabe, 1998, on German; Arvaniti et al., 2017, on Polish) or tone delinking (Grice, 1995, on English), and the temporal shifting of tonal targets (Arvaniti & Ladd, 2009, on Greek). Additionally, tonal crowding can lead to segment elongation to accommodate the tune (e.g., Arvaniti & Ladd, 2009, on Greek; Arvaniti et al., 2017, on Polish). Contra the hypothesis advanced by Grabe (1995) that each language employs only one method of tonal crowding resolution, all of the above adjustments are possible in all languages (for empirical evidence against the idea of a single tonal crowding resolution strategy per language, see Grabe et al., 2000, on varieties of English; Prieto & Ortega-Llebaria, 2009, on Catalan and Spanish; Arvaniti et al., 2017, on Polish; Rathcke, 2017, on Russian; Roettger, 2017, on Tashlhiyt Berber).

The effects of tonal crowding can be pervasive. For instance, Torreira & Grice (2018) found that certain Spanish tunes requiring more than one prosodic word are reduced in different ways by Spanish natives and Italian learners (who appear to use their own prosodic system) when only one prosodic word is available. The reduction involves association of tones to different parts of the metrical structure, such that, e.g., a H tone associated with the nuclear syllable in the longer version is an edge tone in the shorter one. Arvaniti et al. (2017), on the other hand, found that the accentual rise of one of the two Polish tunes they investigated shows compression, while the other shows truncation. Based on this difference, they argued that the accent of the former tune is best represented with an initial L tone (LH*), since that tone cannot be eliminated under tonal pressure, but the latter is simply a H*, since its initial rise can be elided. Thus, Arvaniti et al. (2017), proposed that tonal behaviour under crowding can be used as a criterion for tonehood.

Further, tonal alignment differs across languages and varieties of the same language. For example, many studies have reported looser alignment for some tonal events than a strict interpretation of the SAH would suggest (Welby and Loevenbruck, 2005, on French; Smiljanić, 2006, on Serbian and Croatian; Arvaniti & Ladd, 2009, on the Greek L- phrase

accent; Roettger et al., 2013 on Tashlhiyt Berber). Studies have also reported alignment differences across varieties of the same language (e.g., Grabe et al., 2000, and Fletcher, Grabe & Warren, 2005, on varieties of English). Based on their results from pitch accents across German varieties, Atterer & Ladd (2004) argue that details of tonal alignment are mainly a matter of phonetic implementation and not phonologically determined. This is in line with the principled separation between phonetics and phonology that is a cornerstone of AM but also shows that tonal alignment cannot be relied upon as the sole criterion for tonehood.

In short, a wealth of studies on tonal alignment clearly show that a strict interpretation of the SAH as invariant alignment is untenable: tonal targets show amply documented variability (see Arvaniti, 2016, for a review of possible sources). Though variability in intonation is often seen as deeply problematic, especially with respect to tonal alignment, it is predicted by AM assumptions regarding the relationship between abstract phonological representations and phonetic realisation. However, how to handle this variability during analysis remains a challenge for the traditional methods and heuristics espoused by AM. Most importantly, a strict interpretation of the SAH can lead to circularity: underlying tones are said to be expressed as invariant targets but often the sole argument for the presence of an underlying tone is target invariance (see e.g., Arvaniti, 2019, Arvaniti et al., 2024, for discussions). We address newer approaches to these concerns in section 6 and their repercussions for AM in section 7.

A corollary of the focus on tonal targets is that the course of F0 between targets has received relatively little attention, on the assumption that interpolation between targets is automatic and thus of little interest. However, evidence suggests that important phonetic detail may be missed by ignoring interpolation. For example, as shown in Fig. 3, the rise of L*+H in English is concave, while that of L+H* is convex, a difference that cannot be explained by monotonic interpolation. Two approaches have been proposed to address such differences. One involves the assumption that tones have duration rather than being single points in the F0 trajectory (Cole, 2000). If so, the shape difference between L*+H and L+H* is an epiphenomenon of a longer low target for L* in L*+H than in L+H*. The second approach involves different types of interpolation that lead to changes in slope and accent shape and are linked to perception (Barnes et al., 2012). Either way, the issue of interpolation, though still largely a matter of phonetic implementation, is clearly relevant for distinguishing tonal categories, as it affects how they are perceived and thus, it deserves closer study.

The importance of phonetic detail is also supported by perception studies in which either the slope and duration of a rise or fall was manipulated while keeping peak alignment constant or the duration of a peak was manipulated, producing a plateau. D'Imperio (2000) showed that plateaux can shift the perception of a categorical pitch accent contrast in Neapolitan Italian (see also Gili Fivela & D'Imperio, 2010). Niebuhr (2003) modified tonal alignment in conjunction with rise and fall steepness to indicate information status in German, and found that independently of alignment, more 'new' responses were obtained when a fast rise was combined with a slow fall. Knight (2008) showed that in Southern British English, plateaux lead

to the perception of higher and more prominent accents relative to peaks. Kimball & Cole (2016) reached similar conclusions about American English, showing that the effect of the H* shape (plateau or peak) is perceptually relevant, with plateaux being better discriminated than peaks. Finally, Dorokhova and D'Imperio (2019) found that in French, F0 shape is responsible for intonation category perception, independently of both tonal alignment and scaling. Such results indicate that tonal alignment cannot be the only cue to intonation category identification.

Additionally, several studies have shown that pitch span differences can encode pragmatic meanings in several languages in spite of similar tonal alignment (cf. Hirschberg and Ward, 1992). For instance, Savino & Grice (2011) investigated polar questions in Bari Italian and showed that while both neutral and negatively biased questions have the same rising accent, the accent's pitch span is greater (leading to a higher peak) in the latter, a difference listeners use to categorically interpret a question as either neutral or biased. Similarly, Borràs-Comes et al. (2014) found that in Catalan, the same rising accent can be used to indicate information focus and corrective focus (in statements), as well as counter-expectational questions, depending on the accent's pitch excursion (though only the difference between statements and questions was interpreted categorically). More recently, Orrico & D'Imperio (2020) reported that the interpretation of positive bias in canonical questions is affected by pitch span in Salerno Italian, with wider span related to more positive bias (though the results were affected by individual variability).

Studies like those reviewed above indicate that the stability of tonal alignment in production is not as strict as once thought, and its role in perception is moderated by additional cues, including differences in interpolation, pitch span, and the shape of the tonal targets themselves.

Finally, the stability of targets presupposes that production is minimally different among individuals. However, research shows that speakers may use different strategies to encode intonational categories (see Cangemi, Krüger and Grice, 2015, for an overview). Niebuhr et al. (2011) examined the production of the H* ~ H + L* contrast of German and the L+H* ~ L*+H contrast of Neapolitan Italian and concluded that there are speaker-specific production strategies, with "aligners" – speakers who show stable tonal target alignment – and "shapers" – speakers who encode a category by means of F0 shape that is not stably aligned with the text – being at the two ends of a spectrum. These authors proposed that, in order to maintain a contrast, individuals may consistently produce either the shape or the alignment of a specific accent. Similarly, work on the narrow vs. broad focus distinction in Egyptian Arabic (Cangemi et al., 2016) showed large individual differences, with speakers varying in the number and combination of the parameters they use (scaling and alignment of H and L tones). The variability in this dataset was considerably reduced when a holistic measure was employed additionally to taking shape into account (Bruggeman et al. 2018, see section 6.2.1). More recently, Gryllia et al. (2022) reported a shape-realisation difference in Greek H* L-L%, with both rise-fall and plateau-fall productions of the tune, a difference linked both to task and speaker-specific variation.

To sum up, static temporal features of tonal targets are not the only relevant dimension in tonal event identification; at the production level, different strategies might be adopted by speakers in signalling intonation contrasts. Thus, language, dialect and speaker-specific patterns used in distinguishing intonation contrasts in both production and perception should be further explored.

5.2. Problems with standard methodologies: Annotation

The traditional methodology for investigating the phonetics of intonation, though well established, presents a number of problems, the first of which is that it requires the annotation of melodic categories. This presupposes that the categories are well established in the language under investigation, and that they are easy to locate and identify in speech. However, extensive and careful research is first required to establish the system (Jun & Fletcher, 2014). Even when a system is reasonably well defined, annotation cannot be undertaken without extensive prior training and patience. Further, annotation is time-consuming, making it difficult to collect and annotate large corpora, especially of unscripted speech, where canonical descriptions of tonal events are not always easy to reconcile with the speech signal. Additionally, annotation requires a reliability check, typically performed by having a subset of the data annotated by a second researcher. If annotation is sped-up by having several annotators work on parts of the same corpus, difficult cases need to be discussed, and a measure of inter-transcriber consistency must be reached, but high agreement is not always easy to achieve (on MAE_ToBI, see Pitrelli, Beckman & Hirschberg, 1994, Syrdal and McGory, 2000, and Yoon et al., 2004; on German, see Grice et al., 1996).

Two issues are associated with potential disagreements: locating and identifying tonal events and determining their tonal targets. The first is perhaps the most challenging, as it requires extensive auditory training. In this step, the annotator first must determine if an event has occurred. Since AM does not automatically assume that all pitch undulations are phonologically relevant, this is not a straightforward task and can be particularly challenging with respect to accents: annotators must first determine if a word is accented and subsequently the type of accent involved. As we know from relevant studies, inter-annotator agreement can be low; e.g., Yoon et al. (2004) compared the data from two annotators working on English, and reported good agreement (Fleiss' kappa of 0.75) regarding the presence vs. absence of an accent, but only moderate agreement (Fleiss' kappa of 0.51) for accent type.

The second step in traditional annotation is to determine the tonal targets involved in the realisation of a given category. Even after extensive training with an established framework that includes training materials – such as MAE_ToBI (Brugos, Shattuck-Hufnagel, & Veilleux, 2006) for American English, and G_ToBI for German (Grice, Baumann & Benz Müller, 2005) – the practice of locating tonal targets, seen as reflexes of underlying melodic events, is fraught with difficulties. In the first instance, annotators must be sufficiently trained to be able to distinguish between microprosodic perturbations, such as F0 depressions during voiced obstruents, and

genuine maxima and minima. In addition, not all underlying tones are realised as easy to locate F0 minima and maxima. Instead, tones may be realised with some duration, forming plateaux. Finally, transitions from low to high targets and vice versa often result in “elbows”, showing gradual changes; locating an actual target in such contexts can be difficult, though solutions have been proposed over the years (see D'Imperio, 2000, for a geometrical determination of elbows, and Giudice et al., 2007, for a comparison of elbow-locating methods).

Finally, it should not be forgotten that focusing on targets ignores the fact that the actual minima and maxima are the output of a pitch tracking algorithm. This makes automatic target location challenging though not impossible; e.g., AuToBI (Rosenberg, 2010), a model for automatic annotation based on AM, offers reasonable results for American English and can be used as a first step to annotation followed by manual correction. Critically, however, the fact that peaks and troughs located using a tracking algorithm are not always consistent makes it clear that phonetic targets should not be seen as an absolute determiner of the location of underlying tones, let alone be treated as the sole criterion for determining the categories of an intonation system and their phonological representation.

6. Newer methodologies and findings

As explained in section 5, recent research has uncovered issues with the standard methodologies adopted in AM and some of the assumptions behind its phonetic implementation component. As a result, new research questions and methodologies are being explored. In this section, we outline some promising new methodologies, though we stress that the gamut of all new approaches is too large to do justice to them all. We address first newer approaches that deal with the annotation of spoken data; this is followed by a discussion of several models, including models that take into consideration perceptual aspects of F0 realisation (section 6.2.1), models that focus on F0 curves (section 6.2.2 and 6.2.3 on Functional Principal Component Analysis and GAMMs, respectively), dynamical models that aim to address the origins of intonation categories (section 6.2.4), and studies on the interaction between segments and tunes (section 6.2.5). All these approaches aim to understand the nature of intonation, on one hand, and document and tackle its variability, on the other.

6.1. Annotation

The most widely used system for prosodic annotation based on AM principles is Tones and Break Indices, or ToBI (see Jun 2022, for an overview, and Ladd, 2022, for a critique). Variations of this system have been extensively used as a tool for annotating data and developing analyses of prosodic systems for scores of languages (see inter alia, Jun, 2005a, Jun, 2014a, Gussenhoven & Chen, 2020). As noted in section 5.2, however, annotation is a challenging task. The challenges disproportionately apply to languages for which ready-made AM-based schemas are not available. The difficulties mean that our knowledge of intonation systems remains limited, with fewer than 10% of the world's languages having full

descriptions so far. Thus, one goal of newer research has been the development of annotation methods that are less time-consuming and require less training.

6.1.1. Rapid prosody transcription and the concept of prominence in intonation

One method that has become increasingly popular is Rapid Prosody Transcription or RPT (Cole, Mo, & Hasegawa-Johnson, 2010). The aim of RPT is to provide an understanding of prosodic structure essentials, as captured by untrained participant responses. In RPT, participants listen to utterances and mark on unpunctuated transcripts phrase boundaries and the words that sound prominent to them. Boundary strength and word prominence are determined by the extent to which listeners agree in selecting them; e.g., prominence is operationalised as the percentage of participants marking a word as prominent, known as a *p*-score. Thus, RPT, unlike standard methods, does not require consensus between annotators and can be used to explore disagreements among them (cf. Orrico et al., 2025).

The output of RPT is analysed using a large number of potential factors and determining post-hoc whether they are predictive of *p*-scores (e.g., Cole et al., 2010a, 2010b). In more recent studies, instead of *p*-scores, binary responses (words marked as prominent or not) have been used as the dependent variable in Generalised Linear Mixed Models (Baumann & Winter, 2018; Cole et al., 2019; Bishop et al., 2020; Arvaniti et al., 2022; Im et al., 2023; Orrico et al. 2023, 2024, 2025). This method also allows for the analysis of random slopes, used to study individual variability in responses (Baumann & Winter, 2018; Orrico et al. 2023, 2025).

RPT has been used on a number of languages (see, e.g., Baumann & Winter, 2018, on German; Bishop, 2016, Bishop, Kuo & Kim, 2020, and Im, Cole & Baumann, 2023, on American English; Riesberg et al., 2020, on Papuan Malay and German; Calhoun, Wollum & Va'ai, 2021, on Samoan; Orrico et al. 2023, on British English; Orrico et al., 2024, on Greek). By and large, research shows that phrasal boundaries are detected relatively easily by untrained participants even when they are not familiar with the language they annotate (Cole & Shattuck-Hufnagel, 2016). Prominence, on the other hand, has been more problematic, potentially because the concept itself is not rigorously defined and its cues (and cue weighting) are language specific and subject to individual variation based on listener cognitive style (see Ladd & Arvaniti, 2023, for a critical overview; see Bishop, Kuo & Kim, 2020, and Orrico et al., 2025, for the role of individual variation in prominence judgments).

RPT rests on the assumption that words that stand out relative to others will be marked as prominent by listeners and that these words will be those that are metrically strong and potentially accented (though the methodology cannot distinguish between the last two options). Perceived prominence, investigated using a visual analogue scale as well as with RPT, has also been linked to periodic energy mass, a measure related to metrical strength, incorporating periodic energy and duration (Albert, Cangemi & Grice, 2018; Albert et al., 2020; Albert & Nicenboim, 2022; Lialiou et al. 2021; Grice et al., 2025; see section 6.2.1; on the perceptual integration of pitch and duration, see Steffman and Jun, 2019, and Steffman & Zhang, 2023).

Though prominence and its investigation using RPT or visual analogue scales are becoming increasingly popular, it is also fair to note that the connection between prominence assessment and phonological categories such as pitch accents is not always straightforward. Research has shown that prominence responses in RPT studies vary across participants (Roy, Cole & Mahrt, 2017) and are influenced by several parameters, many of which are unrelated to intonational or, more broadly, prosodic information; such parameters include word frequency, part of speech, previous mentions and the like (e.g., Baumann & Winter, 2018; Cole & Baumann, 2023; Im et al., 2023). Further, various studies have shown that responses to RPT are susceptible to the instructions used; e.g., Cole, Mahrt, & Hualde (2014) and Cole et al. (2019) found that responses differed depending on whether the instructions emphasised semantic or acoustic factors. More recently, Orrico et al. (2024) asked Greek participants to choose either the most prominent word in each utterance or mark all the words they found prominent; both groups consistently selected the word with the nuclear accent as most prominent, but the latter group relied on acoustic parameters, such as maximum F0, to a larger extent. Orrico et al. (2024) argue that the difference is due to task demands (as encoded in the instructions) and that, therefore, RPT responses cannot be treated as direct reflexes of prominence (or of its connection to stress, phrasing and accentuation). Moreover, the translation of instructions into the language under investigation can lead to differences that have to do with terms related to prominence not being comparable across languages (on issues created by an understanding of the task and by different task instructions, see Cole et al., 2014, Cole et al., 2019, and Orrico et al., 2024).

Additionally, studies relying on speakers of different languages suggest that prominence is less signal-driven than originally thought, in that RPT responses are not as straightforwardly connected to acoustic aspects of the signal as the first studies, which were based on Germanic languages, suggested. For instance, Riesberg et al. (2020) conducted RPT using German and Papuan Malay stimuli which were tested with native speakers of both languages. The Papuan Malay speakers had more difficulty with the task and disagreed more with one another on their native language than on German. Similarly, Calhoun et al. (2021) report more conclusive results from their English than their Samoan participants, while Smith (2011) found that French participants had, like Papua Malay speakers, difficulties with discerning prominent words but very good agreement in phrasal boundary detection. Similarly, according to Cole et al. (2019), Spanish and French listeners' responses were more influenced than those of English participants by the two sets of instructions used in their study, one emphasising phonetic, the other pragmatic factors. Perhaps more strikingly, Orrico et al. (2024) report that Greek participants consistently chose as most prominent words accented with L*, provided the L* accent was focal, even though these words were not as acoustically salient as other items in the same utterance; in contrast, H* focal accents were not considered particularly prominent by the same listeners. In short, L*-accented items were considered more prominent both syntagmatically (in relation to either items in the same utterance) and paradigmatically (when compared to other focal accents). These results indicate that the finding of Baumann

& Winter (2018) that L^* is the least prominent accent while H^* is among the most prominent must be language-specific. A way to understand such differences is offered by Ladd & Arvaniti (2023), who argue that an understanding of prominence is more likely to be reached if relative metrical strength is considered in addition to phonetic parameters.

Despite the above limitations, it is fair to say that RPT is a robust paradigm in that aggregate results are consistent despite substantial differences among individual participant responses (Cole, Mahrt & Roy, 2017; Arvaniti et al., 2022; Orrico et al., 2025). Thus, RPT can be a helpful exploratory tool, provided native speaker intuitions and phonology are taken into consideration and that RPT findings are not seen as the definitive answer to questions that are essentially phonological, such as whether a language has stress or pitch accents. If the speakers of a given language show low agreement in their RPT responses, as did the French and Papuan Malay speakers discussed above, this should be taken as an indication that the language may lack stress (i.e., differences in metrical strength), rather than as an indication that the language in question is an aberrant case or a language with weak prominence exponents. Variable responses may also mean that the prosodic system does not fit standard assumptions that largely derive from Western European languages. If such findings are the outcome of RPT, then a closer look at the language's prosodic system is warranted.

6.1.2. Prosodic annotation and systematic phonetics

Since tonal invariance is elusive and targets are more variable than expected, several researchers have attempted to tackle variability by treating AM representations as transcriptions, i.e., as systems of annotation intended to capture systematic phonetic differences (what Arvaniti & Ladd, 2009, dubbed "lawful variability"; for a review of the role of systematic phonetics in phonology, see Ladd, 2014, chap. 2). This approach has taken many forms, some of which we review in this section.

One such approach is taken in analyses that expand the use of secondary association (briefly discussed in section 3.2), a mechanism first proposed by Pierrehumbert & Beckman (1988) to account for a tonal spreading pattern observed in Japanese. Secondary association was used by Gussenhoven (2000), which posited that boundary tones in Roermond Dutch are phonologically aligned with both the right edge of the phrase and an additional leftmost position. Grice et al. (2000) also relied on secondary association to account for the spreading of phrase accents in various tunes of English, Greek, Cypriot, Hungarian, and Romanian: the phrase accents are analysed as having a primary association to a phrasal boundary and a secondary association to a stressed syllable. Though secondary association was largely used as a mechanism for capturing discrete changes in tunes related to linguistic factors, it has also been used to capture differences in the tonal alignment of pitch accents; e.g., Prieto, D'Imperio & Gili Fivela (2005), used secondary association to capture a distinction between rising accents ($L+H^*$), which have a delayed peak in prenuclear position in Catalan statements with broad focus but an earlier peak when occurring in the same position in imperatives; the earlier peak is accounted for by means of a secondary association. Whether it is necessary to capture this

type of detail in phonological representations has been questioned by Arvaniti, Ladd & Mennen (2006b), who argued that such differences are more economically analysed as matters of phonetic implementation rather than as phonological distinctions.

A more widely adopted approach to capture phonological distinctions and their phonetic variants is to create both phonological and phonetic levels of annotation. This applies, for instance, in the annotation systems of Korean (Jun, 2005b) and British English as represented in the IViE system (Grabe, 2001). These systems include a tier for phonological annotation broadly based on AM principles plus a tier of phonetic annotation, the details of which vary by system. In Jun (2005b), this level is akin to a broad phonetic transcription, i.e., a way to discretise the variation found in the pitch contours by means of allotones. In contrast, the phonetic annotation in IViE is closer to a narrow phonetic transcription, capturing the surface shape and timing of the F_0 contour relative to the stressed syllable.

The idea of including a level of systematic phonetics in annotation has been incorporated into the proposal for an International Prosodic Alphabet or IPrA by Hualde & Prieto (2016). Specifically, these authors proposed a system of schematic representations of possible accents together with suitable labels that can be used for all languages; e.g., rising accents with peak delay should be annotated as $L+<H^*$ cross-linguistically. The system is inspired by the International Phonetic Alphabet (IPA) and is meant to be "broad enough to encompass differences in phonetic realisation". IPrA, however, deviates from the principles of the IPA in some important respects (see Arvaniti, 2016, for a critique of IPrA). First, the International Phonetic Association recognises that the number of phonetic symbols is finite and thus one cannot (and should not attempt to) fully capture all possible phonetic variation (IPA, 1999). Additionally, the IPA recommends that researchers focus on phonological (i.e., contrastive) representations accompanied by detailed phonetic descriptions, as this facilitates cross-linguistic comparisons by keeping categories broad. For instance, a symbol such as [s] may be used for the voiceless coronal sibilant of both British English and Standard Greek, accompanied by a description such as that the former is alveolar and can be either laminal or apical (Jones, 1972: 185; Cruttenden, 1994: 170), whereas the latter is a laminal retracted alveolar (Arvaniti, 2007, and references therein). In contrast, IPrA symbols are meant to have a very strictly defined phonetic realisation requiring the use of multiple diacritics: e.g., $L+H^*$ is to be used for accents whose peak is reached at the end of the accented vowel and $L+<H^*$ for accents with delayed peak. Instead, the IPA recommendation would be to use one symbol for both and reserve diacritics only if the accents are used contrastively in the same system or a narrow phonetic transcription is the goal. It is possible that a more flexible system, along the lines of the IPA, may be useful for the latter purpose, though in our view, newer modelling methodologies, such as those discussed in section 6.2, may largely supersede the need for a level of systematic phonetic transcription of intonation.

Another methodology developed to curb the need for extensive annotation training is PoLAR (Ahn et al., 2019a; Ahn et al., 2019b). PoLAR draws upon the AM tradition (though not exclusively so) and aims to be a system suitable for novice

annotators with minimal training. Like ToBI, annotation consists of tiers, described below and illustrated in Fig. 7.

- Prosodic Structure* is a tier used to indicate the presence of a perceived prosodic prominence and prosodic phrase boundary;
- Pitch Points* is a tier for labelling F0 “turning points” (peaks, valleys, and plateaux edges);
- Scaled Levels* has a 1-to-1 relationship with the Points tier but records a numerical value that corresponds to pitch range operationalised as a number of levels;
- Range Domains* is a tier in which annotators define stretches of the utterance that have their own local pitch range.

The aim of the tiers used in PoLAR is to capture both phonological structure, in the Prosodic Structure tier, and phonetic detail, in the Levels and Ranges tiers (in a manner similar to the IViE phonetic annotation), and to do so with less training than traditional AM labels and in an easier manner. To achieve this, the structure of PoLAR allows for the information of some tiers to be easily derived from others (e.g., the Levels tier can be automatically derived from the Pitch Points and the Range Domains tiers). PoLAR is also reminiscent of RPT, in that labels in the Prosodic Structure tier aim to reflect the presence of prominences and boundaries, rather than specific pitch accents and edge tones.

Similar assumptions are behind the RaP (Rhythm and Pitch) system (Dilley et al., 2006), which is based primarily on annotators’ perceptual impressions of prosodic events. RaP includes a rhythm tier for labelling metrical prominences and phrasal boundaries, and a tonal index tier for labelling tonal information. In the rhythm tier, annotators mark strong and weak metrical beats, while in the tonal index tier the labels H*, L* and E* are used. According to the developers, the separation of strong beats from accented elements, a practice similar to the IViE system (Grabe, 2001), is helpful during annotation, and it can certainly be an advantage in distinguishing metrical strength from accentuation. Nevertheless, the developers’ own comparison with MAE_ToBI did not reveal a substantial difference in inter-annotator agreement between

the two systems, except in terms of phrasal boundaries (Dilley et al., 2006; Breen et al., 2012).

The annotation issue and the aforementioned approaches are related to the expectation that all underlying tones will be fully and clearly realised in production. Inevitably, this does not always obtain. As with other aspects of speech, disfluencies of various types are possible with intonation, while targets can be elided or undershot for several reasons, such as tonal crowding, melodic context, speaking rate, and speech clarity (for a discussion, see 5.1, and Arvaniti, 2016). Consequently, an expectation that each underlying tonal category will be fully produced in an identical manner by all speakers in all contexts and will thus always remain distinct from all other categories inevitably runs into problems. Positing a level of transcription akin to systematic phonetics is an attempt to discretise the continuous variation that is inevitable in speech. Though this stems from a need to handle variation during analysis, it often leads to more annotation conundrums, as analysts try to shoe-horn all tokens of a given abstract category into a small number of “allotones”. Positing increasingly fine-grained, yet discrete categories, does not solve this problem. This has been convincingly demonstrated by Ladd (2014, chapter 2) on the basis of examples from segmental phonetics, leading him to argue for solutions that do not include a level of systematic phonetics in speech analysis. Arvaniti (2016), and Arvaniti et al. (2024) have argued for a similar approach to the relationship between the phonology and phonetics of intonation as well. Now, the attested presence of fuzzy category boundaries and category overlap in intonation (as in segmentals) does not preclude the possibility of non-overlapping variants. Such a case is reported by Jun (1996) with respect to Korean AP-edge tones which, as Jun shows, have discrete, non-overlapping variants differing in pitch height. It is possible that since the scaling differences reported in Jun (1996) are induced by specific segments, they act as a cue to segment identity (as a type of phonologised microprosody). Thus, it may be important for speakers to keep such allotones distinct, something that may not apply to other cases, such as high

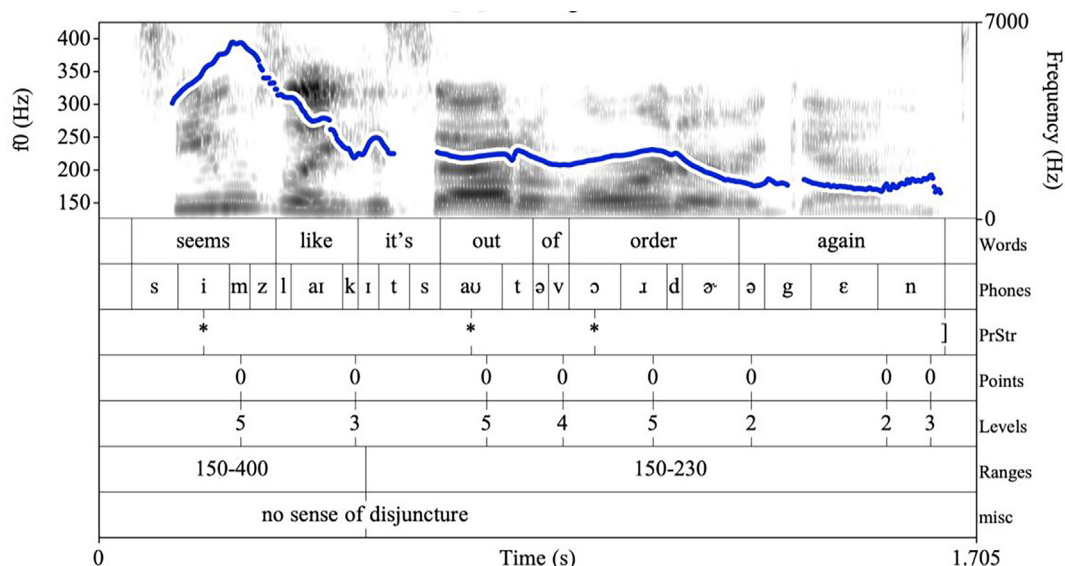


Fig. 7. An illustration of PoLAR (from Ahn et al., 2019b).

pitch accents with early and late peaks reported by Prieto (2015) for Catalan. Either way, the presence of discrete variants as opposed to gradience may be case-specific and as such it remains a matter of empirical investigation.

6.1.3. Separating phonetic from pragmatic criteria

A different approach to annotation has been recently advocated by Kim et al. (2024) in their study on the $H^* \sim L+H^*$ accentual contrast in British English. Traditionally, annotation is performed by considering visual representations of speech (waveforms, spectrograms and pitch tracks), while listening to the audio. In practice this means that tonal categories are classified using multiple phonetic, pragmatic, and contextual criteria (see Cangemi & Grice, 2016, for discussion). This synthetic approach is problematic, as individuals vary in how they weigh different cues in both production (e.g., Hu & Arvaniti, 2024) and perception (e.g., Orrico et al., 2025). As may be expected, this also applies to annotators. Thus, the combination of individual variability in study participant production and annotator response to it can lead to significant disagreements during the annotation process.

To avoid this problem, Kim et al. (2024) categorised English high-pitched and rising accents (typically annotated as H^* and $L+H^*$, respectively) based on their F0 trajectory but without considering other phonetic aspects or the accents' function. Additionally, they tested a specific hypothesis about the pragmatics of H^* and $L+H^*$, namely that H^* is used primarily to encode new information, while $L+H^*$ is used for contrast and correction (Pierrehumbert & Hirschberg, 1990). The pragmatic annotation was done using written transcripts and without recourse to (silent) prosody; e.g., if a speaker said *I've got purple lavender* in response to *Have you got a purple aubergine right there?*, *lavender* was considered contrastive, independently of how the speaker produced it or how the annotator thought they would have produced it themselves. In the following stage of analysis, the pragmatic labels were combined with the phonetic labels. Separating the two processes allowed Kim et al. (2024) to reach good annotator agreement (Fleiss kappa of 0.84, C.I. = 0.78 – 0.89 for phonetics, and 0.72, C.I. = 0.62–0.82, for pragmatics). Crucially, the process allowed them to investigate separately the shape and pragmatics of H^* and $L+H^*$ and conclude that phonetically, British English has two accent types (fall and rise-fall, or H^* and $L+H^*$, respectively) and that both are used to encode new, corrective, and contrastive information. Had the annotation followed traditional guidelines, it would have been difficult to understand if the acknowledged difficulty with H^* and $L+H^*$ was due to their forming a phonetic continuum (in terms of F0 shape) or being used interchangeably for a variety of pragmatic functions. These findings allowed Kim et al. (2024) to reject the pragmatics-based hypothesis they set out to investigate. Though this method is not suitable for all tonal contrasts, it can be a useful first step towards an analytical approach to intonation and its annotation.

6.2. Modelling production and perception

6.2.1. Acoustic methodologies

Analyses of intonation contours based on the acoustic signal rely on the estimated F0 trajectory. As pointed out in

section 6.1, the landmarks extracted traditionally emphasised positions of local maxima and minima along the F0 curve. While such turning points are evidently useful in characterising the F0 trajectory, they are arguably not the positions of the tonal targets (i.e., they are not necessarily privileged in perception), as they can occur outside the syllabic nuclei, which are optimal for bearing pitch information in the speech signal. Thus, while the F0 peak of a $L+H^*$ pitch accent can occur on an intervocalic consonant, the F0 rise is timed to occur within a syllabic nucleus. It is arguably the rise, rather than the F0 peak, that is important for perception. The onglide measure developed by Ritter & Grice (2015) incorporates some of the F0 rise or fall, but still relies on a tonal target (i.e., a turning point) as the endpoint of the F0 curve that is taken into account for this measure; see section 6.2.4 for more details on the onglide.

A notable attempt to go beyond F0 turning points was made by Barnes et al. (2012), who proposed a new measurement, the Tonal Center of Gravity (TCoG). TCoG represents an average point in time, weighted by F0. Instead of simply tracking turning points along the F0 curve, the TCoG takes into account the shape of the curve within a given rise-fall or fall-rise trajectory. Thus, the TCoG values change along with changes in rise and fall shape, to reflect the position of the tonal target relative to L and H peak positions by accounting for asymmetries in the F0 contour, which can lead to perceptual shifts in pitch accent identification. In fact, perceptual judgments of accents may be shifted along with changes in the shape of the F0 contour even if the alignment of the F0 peak remains unchanged (D'Imperio, 2000; D'Imperio et al., 2000, for plateau effects). Barnes et al. (2012) used the TCoG as an alternative to F0 peaks in tonal alignment and scaling measurements, deriving different values for domed (convex) vs. scooped (concave) rises (where domed rises shift the TCoG before the F0 peak, and scooped rises shift it after). TCoG has also since been used to describe categorical distinctions in languages other than English, such as Egyptian Arabic (Bruggeman et al., 2018).

While the TCoG can be considered as an incremental step away from the notion of F0 turning points, a more radical proposal that incorporates the TCoG suggests dispensing with F0 turning points altogether. The open toolbox for Prosodic Analysis with Periodic Energy (ProPer; Albert et al., 2020; Albert, 2023:141–158) utilises the interaction of the F0 time series with the corresponding *periodic energy*, which reflects the acoustic intensity of the periodic portions in the signal (i.e., the pitch-inducing aspects of the signal, excluding aperiodic *noise*; see Oxenham, 2012). Acoustically, periodic energy is based on distinctions between regularly and randomly recurring patterns, using methods such as auto-correlation (Boersma, 1993). ProPer therefore utilises the interaction of two acoustic dimensions that are assumed to reflect the *quality* (F0) and *quantity* (periodic energy) of perceived pitch. Moreover, the periodic energy curve is similar to the amplitude envelope of filtered audio signals that capture syllable-size fluctuations in speech (e.g., Tilsen & Arvaniti, 2013; Räsänen et al., 2018).

Rather than describing the F0 contour from the perspective of landmarks along the F0 trajectory itself, ProPer describes F0 contours from within syllable-size windows, as determined by the shape of the periodic energy curve (with dips in the curve corresponding to syllable margins). One metric that ProPer

uses to characterise pitch contours, referred to as Synchrony (Cangemi et al., 2019), adopts the TCoG (in its later version: Barnes et al., 2021). In each syllabic interval, for which the TCoG is obtained, Synchrony also extracts a corresponding landmark that is derived from the periodic energy curve alone: the centre of periodic energy mass (CoM). CoM is based on a weighted average function that takes periodic energy as its weighting term for time, and corresponds to the acoustic centre of the syllable. Crucially, the distance between TCoG and CoM is informative with respect to the F0 shape within a given interval. Synchrony computes these temporal asynchronies such that positive values indicate a mostly rising trend, negative values indicate a mostly falling trend, and increasing values in either direction indicate increasingly steep trends. Synchrony has been used to analyse intonation patterns in Italian (Cangemi et al., 2019; Sbranna et al., 2023), German (Roessig et al., 2022), English (Jeon & Nichols, 2022) and Brazilian Portuguese (Silveira et al., 2025).

A more recent proposal to provide configurational characterisations of F0 contours based on production models comes from Iskarous et al. (2024). Their model, based on dynamical systems theory, uses differential equations to study the relationship between F0 and its derivatives (the first derivative, F0 velocity, being of special interest, given its ability to capture important aspects of the F0 curve with respect to distinct pitch accents). Iskarous et al. (2024) devised a model that represents the pitch-producing motor-system with emphasis on its F0 lowering and raising forces. We return to dynamical systems in section 6.2.4.

6.2.2. Functional principal component analysis

Additional recent developments include the use of Functional Principal Component Analysis (FPCA) (Ramsay & Silverman, 2005). FPCA was applied by Gubian, Cangemi & Boves (2010) for F0 synthesis, and by Gubian, Torreira & Boves (2015) to answer questions related to Spanish segmental phonology by investigating tonal alignment in diphthongs and vowels in hiatus. It has since been used in studies on intonation, such as Asano & Gubian (2018), and Arvaniti et al. (2024) (see also Grabe, Kochanski, & Coleman, 2007, Chen & Boves, 2018, Baltazani, Przedlacka & Coleman, 2019, and Baltazani et al., 2022, for uses of Functional Data Analysis in the study of intonation).

FPCA is a method for modelling curves (in the case of intonation, F0 curves) in terms of the main modes (or *principal components*) of variation in the data. In FPCA, each curve in the dataset under analysis is represented as a function that consists of the mean curve and a number of principal component curves (PCs), each multiplied by a coefficient (or *score*) which represents the contribution of this PC to the shape of a specific dataset curve (for detailed accounts of FPCA using F0 curves, see Gubian et al., 2015; see Arvaniti et al., 2024, for a non-technical introduction and application of FPCA to intonation).

The use of FPCA is motivated by the need to move beyond the investigation of tonal targets and alignment and is inspired by newer methods such as TCoG, discussed in 6.2.1. What these methods have in common is that they use as their input stretches of the F0 contour rather than specific points. The motivation behind FPCA in particular is to get a handle on

the variability discussed in section 5.1. First, FPCA allows for significant dimension reduction thereby minimising data noise. An additional advantage is that FPCA is data-driven and thus, for scripted data in particular, no annotation is necessary, other than determining the stretch of F0 over which FPCA is to be performed. Further, because principal components are by definition independent dimensions along which the data vary, FPCA can be used to study tune compositionality: e.g., if a particular configuration, such as a rise-fall, is reflected in one principal component (PC), then it is reasonable to assume that it forms a melodic unit, whereas if the rise and fall are reflected in different PCs, then it is more likely they are independent events (how the unit or units should be phonologically represented is a separate matter).

Most importantly perhaps, the PC scores that are part of the FPCA output can also be used for statistical modelling. Such modelling can be used to determine the extent to which specific modes of variation reflect linguistic distinctions; as such, it can help distinguish linguistic variation, such as systematic changes that apply to a tonal category due to linguistic context, from general variability (cf. Bürki, 2018). Thus, the statistical modelling of FPCA scores addresses the need to separate general variability from linguistically-determined variation (Arvaniti et al., 2024). As an illustration of these two points, the FPCA reported in Arvaniti et al. (2024), which dealt with three pitch accents of Greek (H*, L+H*, and H*+L), showed that two F0 dimensions – F0 height and shape – are sufficient to distinguish the three accents: H* and H*+L are falls with lower and higher scaling respectively, while L+H* varies in scaling but has a distinct scooped shape. These two dimensions are reflected in the first two PCs in the analysis of Arvaniti et al. (2024), indicating that tonal scaling and shape are independent of one another and together define the three accents. (The same does not apply to traditional scaling and alignment however: the analysis shows that they covary as a result of specific tonal pressures, including tonal crowding.) PC3, on the other hand, though it accounts for nearly 9% of curve variance in the data examined by Arvaniti et al. (2024), is not linguistically relevant: a linear mixed model on the PC3 scores with accent as the fixed factor shows that the curve variance reflected in PC3 is not related to accent (marginal R^2 was just 0.05) but to item-related variation (conditional R^2 was 0.55). This type of analysis can help disentangle dimensions of variation that should be reflected in the intonation system – the need to posit three distinct accents, in this case – and incidental variability that should not be a cause for concern and does not require phonological representation.

Finally, FPCA can be used to examine together more than one signal in what Asano & Gubian (2018) dubbed 2D-FPCA. This feature can be very useful in examining whether aspects of tonal realisation co-vary with other phonetic parameters, and if so, in what way. For example, Arvaniti et al. (2024) showed that in Greek L+H*, F0 and duration covary in a way that can be construed as cue trading: less scooped and lower scaled L+H*s are co-produced with longer duration, suggesting that duration is used to compensate for the lack of prototypicality in the F0 realisation of this accent (see also section 6.2.5).

6.2.3. Generalised additive mixed models (GAMMs)

Another method for examining F0 curves is Generalised Additive (Mixed) Models (Hastie and Tibshirani, 1986; Wood, 2017). Recent articles that used GA(M)Ms to study intonation include Zahner et al. (2019) and Barnes et al. (2021), while Sóskuthy (2021) provides a hands-on explanation on how to work with GA(M)Ms.

A GAM is an extension of the generalised linear model where the linear component is replaced by the sum of smooth functions (and the contribution of each non-linear function on the relationship between each predictor and the response is summed or added together). Unlike standard linear regression models where the relationship between the independent variable and the response variable is linear (i.e., a change in the independent variable results in a proportional change in the response variable), GAM(M)s do not make such an assumption about the data and can be used for response variables with a Gaussian, Poisson or binomial distribution. Since linguistic contrasts under investigation in intonation research often involve identifying linguistically meaningful F0 turning points and their (relative) location, scaling, and, increasingly, the shape of the F0 curvature, GA(M)Ms are well suited for intonation research.

An additional reason why GA(M)Ms are advantageous for research on intonation is that they are additive models – i.e., the effect of each predictor on the response variable can be examined with the “all-else-being-equal” situation. This makes GA(M)Ms highly interpretable and suitable for research involving multiple independent variables, as one can easily plot an estimated component function for each individual variable while assuming the others are held constant. This is in addition to the interpretability of any generalised linear model. (Though, it should also be noted that adding and configuring interaction terms between variables is a more involved process in GA(M)Ms than other types of modelling.) While some other types of analysis, such as random forest or boosted trees, are better at modelling complex relationships between the independent and response variables, those analyses typically lack the level of interpretability that GA(M)Ms have. Another possible advantage of GA(M)Ms is that they do not require a separation of the analysis of curves from the statistical modelling, which can be particularly useful for addressing some research questions.

A limitation of GA(M)Ms is that they require a priori categorisation of the tunes under investigation. Thus, depending on the research question, the use of GA(M)Ms may require the annotation of intonational categories, which, as discussed, is a time-consuming task that could lead to some circularity (e.g., if the categorisation of tunes is based on F0 shape, then GA(M)Ms will reflect the shape difference that is already built into the categories). To avoid this issue, some researchers use GA(M)Ms together with FPCA to answer different but complementary questions (Kim et al., 2024). In this approach, FPCA can be used to help discover the main parameters along which tonal categories differ. Further, since FPCA is data-driven, its output, if in line with the GA(M)M results, can help alleviate concerns about circularity. GA(M)Ms, on the other hand, can provide information on the changes and location of changes in curve shape. Further, GA(M)Ms can be particularly useful for studies in which the grouping criterion (such as accent categories) is not itself the focus of inquiry (cf. Kim, Tibbs & Arvaniti, 2025).

Finally, GA(M)Ms can be especially helpful in investigating individual variation in production.

6.2.4. Dynamical models of intonation

There is evidence that continuous phonetic and categorical phonological aspects of speech are intricately related. This relationship has been modelled using dynamical systems in terms of articulatory gestures in the framework of Articulatory Phonology (Browman and Goldstein, 1986) and in the parameterisation and planning of speech gestures (Gafos, 2006, Gafos and Benuš, 2006; Roon & Gafos, 2016).

Such models have recently been applied to intonation to account for how variation in continuous aspects of the F0 trajectory is related to categories typically employed in AM for pitch accent types. Roessig et al. (2019) and Roessig (2021) investigated one phonetic parameter, the onglide, which captures the F0 excursion (amount and extent of change) from the pretonic syllable to the annotated tonal target of the accented syllable. In German, variation in the onglide is related to H + L* (falling onglide), H*(shallow rising onglide) and L+H* (steep rising onglide) pitch accents in declaratives with broad, narrow, and contrastive focus respectively. The link between continuous and categorical aspects of the speakers' productions was first made on a smaller but similar dataset with the same focus structures by Grice et al. (2017), where it was found that some speakers modulated F0 more subtly within one category, whereas others modulated F0 to a greater extent, pushing their productions towards a different category. For instance, one speaker produced H* on both broad and contrastive focus, with a small increase in the excursion of the rising onglide in the latter, whereas another speaker produced H* on broad focus, but increased the onglide in contrastive focus to such an extent that it fell into the category of L+H*. Roessig and colleagues, based on the productions of 27 speakers, built an attractor landscape that could differentiate between these three German pitch accents, capturing the three pitch accent types in a continuous dimension. This made it possible to model not only group differences related to the three accents, but also the variation found across (and within) speakers.

Iskarous et al. (2024) modelled an extensive corpus elicited from 130 Mainstream American English speakers, looking at a number of different continuous variables relating to the F0 curve and derived velocity trajectory. Speakers produced different pitch accent types including H*, L+H*, L*+H and L*, followed by H-H%, H-L%, L-H% and L-L% edge tone configurations, giving 16 combinations of pitch accent and following edge tones. The data were elicited using a type of imitation task in which participants heard utterances with the target nuclear tune and had to reproduce it with a different sentence; e.g., they heard *He answered Jeremy* and had to produce *They honored Melanie*. Three names were used as the final word in the productions, *Harmony*, *Madelyn* and *Melanie*; only the F0 on the name's first two syllables (which included the accent) was analysed, with the edge tones serving as sources of contextual variation.

The empirical F0 trajectories were described using a number of parameters: level (F0 min for falling and F0 max for rising trajectories), velocity (amount and direction of F0 change across successive time steps), peak velocity, latency (the time

difference between the onset of the accentual rise and the time of peak velocity), and span (min–max or max–min depending on rise/fall). These values agree well with other published studies on these accents (other than L*, for which there is little empirical work).

In Iskarous et al. (2024), dynamical systems were built incrementally. The final system can model differences between L* on the one hand, and the three rising accents, H*, L+H* and L*+H, on the other, by employing a stochastic dynamical system that incorporates noise and can thus account for within-category variability. According to the authors, the properties of this system allow for the categories of MAE pitch accents to arise naturally. Thus, they argue that the system can capture the differences represented in AM by different notations (single vs. bitonal accents, serial ordering of tonal elements, the use of the star notation) as well as important aspects of the phonetic realisation of these accents (e.g., the fact that the later rise of L*+H leads to a higher peak) without having to resort to a phonetic implementation component.

Both the model by Roessig et al. (2019) and Roessig (2021) and the model by Iskarous et al. (2024) allow for both categorical and continuous aspects of intonation to be captured in one model, dispensing with the need to treat phonological and phonetic components of intonation as separate. By employing dynamical systems, these studies achieve the goal set out in Grice et al. (2017) to integrate continuous and categorical aspects of intonation. Iskarous et al. (2024) point out that a difference between their work and that of Roessig and colleagues is that their approach addresses the concerns of Pierrehumbert & Pierrehumbert (1990) who argued that in some dynamical models, “the discrete is *specified*, not *derived*” (cited in Iskarous et al., 2024: 3). In other words, Roessig et al. (2019) and Roessig (2021) use predetermined AM categories to understand their phonetics. Iskarous et al. (2024) show that in their model accentual categories are derived by manipulating the values and range of specific model parameters (and thus the categories need not be postulated in advance). It is now a matter of empirical validation to determine if such a model can account for the effect of boundary tones in American English, and apply to more natural and extended materials both in American English and other languages. If so, this could be a significant breakthrough in intonation research.

6.2.5. Tune-driven adjustments to the text

In situations of tonal crowding, we have already discussed adjustments to the tune (see section 5.1). In addition, the segmental string itself can undergo adjustments to accommodate tunes that would otherwise have too little sonorant material or time to unfold (see Roettger & Grice, 2019, for an overview of cases where the tune “drives” the text).

First, it is potentially possible to realise intonation without sonorant material. Niebuhr (2012) showed that the Centre of Gravity (CoG) of voiceless German fricatives differs, depending on whether the intended contour is rising or falling, with the former leading to a higher CoG; see also Ritter and Roettger (2014) for similar results.

However, fricative pitch is not an optimal way for intonation to be conveyed. Where intonation is crucial for transmitting pragmatic information, a number of languages appear to insert optimal carriers for this tune, namely vocalic elements

with periodic energy (Roettger & Grice, 2019). For instance, in Tashlhiyt Berber, where fricatives, even voiceless ones, can occupy the syllabic nucleus, speakers tend to insert vocalic elements where they are needed to bear meaningful intonation contours, such as those distinguishing statements from echo questions (Grice, Ridouane & Roettger, 2015; Roettger, 2017). Vowel epenthesis under similar circumstances has also been reported for Moroccan Arabic (Dell & Elmedlaoui, 2002), varieties of Portuguese (Frota et al., 2016), and Tunisian Arabic (Hellmuth 2022). The status of these epenthetic elements is not always that of a fully-fledged vowel with a position in metrical structure, while their presence typically forms a probabilistic distribution, such as in Italian, where complex intonation contours engender the insertion of a schwa more often and with a greater duration than simple contours (Grice et al., 2018). Roettger & Grice, 2019 claim that this adjustment plays a role in the typology of phonological systems, as is evident in vocatives, which have statistically more vocalic elements than other modalities (Sóskuthy & Roettger, 2020).

A related mechanism, reduction in vowel devoicing to accommodate high or rising pitch, has also been found (also as a tendency with considerable variation), both in cases of lexical tone and pitch accent (Beckman & Venditti, 2010; Hirayama, 2009), as well as in intonation. With respect to intonation, we see that in Bulgarian, prominent accent types lead to a smaller degree of vowel devoicing than less prominent accents (Andreeva & Koreman, 2003), while in Greek, vowel devoicing is less common with rises than falls (Kaimaki, 2015).

Finally, vowel lengthening to accommodate complex tunes has also been reported for many languages. An example is provided in Fig. 8, which shows two instances of the Polish routine calling tune (Arvaniti et al., 2017). The duration of the monosyllabic name *Jaś* in panel (a) and the trisyllabic name *Marianna* in panel (b) are 550 ms and 650 ms respectively and most of the difference relates to the accented vowel, which is 275 ms long in *Jaś* but only 104 ms in *Marianna*. Additional languages in which vowel lengthening to accommodate the tune has been reported include Portuguese (Frota, 2002, 2014), German (Gilles, 2005), Hungarian (Ladd, 2008: 183), Italian (Grice et al., 2018) and Fataluku (Heston, 2014). Such changes to segmental duration may even enter in a trading relationship with F0, as noted in 6.2.2 (Arvaniti et al., 2024).

In short, evidence from tune-segment interactions indicates a much closer, one could say symbiotic, relationship between tunes and segments which runs counter to the prevailing view of intonation as “suprasegmental”, i.e., as an independent and largely inert element that sits on top of vowels and consonants. It also means that the autosegments in AM may be less independent than previously assumed (see Grice 2022a, b, for discussion).

6.3. Perception

The principles underlying the perception of intonation are still relatively uncharted. Despite conceptual advances, there is not yet a comprehensive model of intonation which includes the interaction between the various functions of acoustic cues to intonation perception.

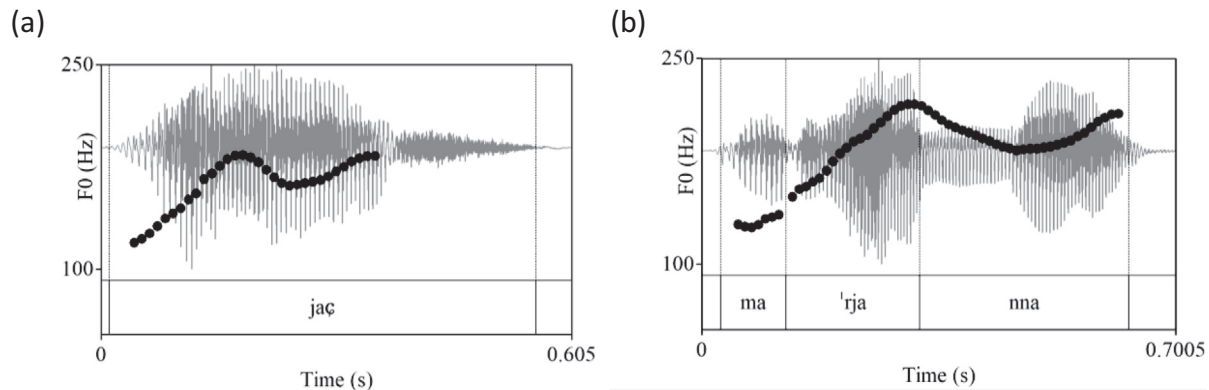


Fig. 8. Waveforms and pitch contours of the Polish routine calling tune demonstrating the effects of the tune on segmental duration under tonal crowding (after Arvaniti et al., 2017); for details, see text.

6.3.1. Determining categories from the link between perception and production: Imitation

A method that is increasingly used to investigate potential phonological contrasts in intonation by linking perception and production is imitation (which was briefly touched upon in section 4). One form of the task involves speakers hearing tunes that change gradiently (in one or multiple dimensions) from one putative category to another, as in Pierrehumbert & Steele (1989). Critically, in imitation, the speakers are required to produce each stimulus immediately after hearing it. The rationale behind the task is that if a stimulus continuum can be split into two because it reflects a contrast, as is the case for English L+H* and L*+H, then most of the stimuli along the continuum will be relatively reliably classified, and subsequently imitated as one or the other of the two categories, resulting in a bimodal distribution of the production values (Pierrehumbert & Steele, 1989). In more recent work, both pitch accents and boundary tones have been successfully imitated in American English (Cole & Shattuck-Hufnagel, 2011). However, certain phonetic parameters appear to be easier to imitate than others. For example, speakers were able to imitate tonal alignment but were not successful in imitating peak height or F0 velocity (Dilley & Heffner, 2013; Tilsen, Burgess and Lantz, 2013). Similar problems are reported by German (2012), a study in which American English speakers imitated the rise-fall of Glasgow declaratives, and by D'Imperio & German (2015), in which Singapore English speakers imitated American English tunes.

Unsurprisingly, the imitation of tonal alignment is sensitive to a speaker's native repertoire, as shown in a study on Bari and Neapolitan Italian (D'Imperio et al., 2014). Bari Italian possesses only one rising pitch accent, L+H* (Grice et al., 2005), whereas Neapolitan has both L+H* and L*+H (D'Imperio, 2002). Bari Italian imitators were able to delay peak alignment so as to achieve a good approximation of the Neapolitan L*+H. Neapolitans were also able to retract the H peak, to imitate the L+H* in Bari questions, though they preserved an alignment difference with the L+H* employed in their statements. A further study with Bari speakers (Savino et al., 2022) involved them imitating the Lecce variety, which has a simple rise in questions. Crucially, the competition from the native repertoire here is between two rising contours, as Bari Italian has a non-

final L* L-H% sequence (a “continuation rise”), which is phonetically distinct from the Lecce question rise (also represented as L* L-H%). Bari imitators were able to approximate the Lecce question rise, although the phonetic realisation was influenced by their native non-final contour in terms of F0 shape.

In an iterative learning experiment, involving imitation of a speaker's own consecutive productions, Braun et al. (2006) showed that speakers gravitate towards a limited number of phonological, linguistically meaningful patterns, which act as “attractors”. Hence, their imitation was based on the ability to reproduce F0 contours that are grammatically meaningful in a specific language variety. Comparable results are reported by Cole et al. (2023) and Steffman et al. (2024). These authors used imitation of various tunes that are predicted for American English by AM and considered also the degree to which the tunes are discriminated in perception. The results suggest that the number of distinctions reliably reproduced and discriminated is much smaller than predicted. Based on these findings, Cole et al. (2023) suggest that the primary pitch accent distinction in American English is between H* and L*, with more fine-grained variation within the H* category possibly reflecting gradient (and meaningful) within-category variation. This primary accentual distinction is coupled with a three-way distinction of high, medium, and low boundary tones (H%, M%, and L%), with no differentiation related to phrase accents.

This is a radical departure from the rich traditional inventory posited for American English (e.g., in Beckman & Pierrehumbert, 1986, and Brugos et al., 2006). Thus, it needs to be carefully considered, as there are possible reasons for these results. First, in Cole et al. (2023), the participants listened to synthesised stimuli and imitation was not direct since they had to reproduce each melody on a different target sentence. It is possible that in this procedure the path between tune identification, phonological tune analysis and imitation was too complex and thus increased variability in realisation. Further, experience suggests that many speakers have difficulty reproducing “on cue” intonational distinctions, even when they spontaneously produce them in their own speech, while others find it difficult to assess pragmatic differences between tunes presented out of context. An additional point may have contributed to the small inventory ensuing from the imitation

study of Cole et al. (2023), namely that the nuclear tune had three syllables upon which to unfold. As mentioned earlier, in situations of tonal crowding there are cases when adjustments are made to a tune. It is fair to note, however, that Cole & Steffman (2023) and Steffman & Cole (2024) report results similar to those of Cole et al. (2023), using the same imitation paradigm but with longer materials. Steffman & Cole (2024) in particular found that some tune distinctions are enhanced when longer materials are used but others are not, and argue that the difference could be used as a criterion for phonological analysis: only distinctions that are *metrically enhanced* (become sharper with longer material) are phonological. The findings of Cole and her colleagues could suggest that the melodic inventory of American English, and potentially the analyses of many other systems, may need revising.

The cognitive ability to imitate intonation contours is not only exploited when explicitly required (e.g., in experimental set-ups as described above). Speakers can also spontaneously imitate the intonation of their interlocutor, a process known as prosodic convergence or accommodation. For example, Savino et al. (2023) show that speakers of Bari Italian, who predominantly produce a rise-fall or rise-fall-rise in polar questions, adjust their intonation in these questions when playing a “guess who” game with Lecce speakers. In such communications, both groups of speakers tend to produce the simple rises typical of Lecce. Similarly, in an interview situation between speakers of Peninsular Spanish and a Majorcan Catalan interviewer, Romera and Elordieta (2013) found that L1 Spanish speakers who had recently moved to Majorca tended to produce polar questions in their native Spanish with a final fall (as in Majorcan Catalan) instead of the final rise typical of their L1. The authors conclude that accommodation may be the mechanism behind changes in intonation systems.

6.3.2. The interaction of production and processing: Aggregate effects and individual variability

We have already shown that the production of intonation categories shows significant variability and have presented early evidence that one of the factors responsible is individual variation. The production results we reviewed earlier suggest that individual speakers may rely more on some features, such as F0 shape, than others, such as alignment and scaling. We also showed that some of these differences could be reduced when taking a different approach to parameterising F0 contours and the sonorant material bearing them. Nonetheless, Cangemi et al. (2015, 2016), Grice et al. (2017), and Gryllia et al. (2023) suggest that even when speakers maintain a contrast, the differences among them may be even less systematic than originally thought.

Further, as we have shown, tonal categories are multidimensional and not all speakers use all possible cues. This is demonstrated in Hu & Arvaniti (2024), who found that most speakers in their sample used not only F0 differences, but at least one of duration, amplitude, or voice quality to differentiate H*, L+H* and H*+L in Greek. Why some speakers systematically use more cues than others, and whether there is a structure to the combinatorial possibilities, remains an open question.

Additionally, the findings of Cangemi & Grice (2016) suggest that the relationship between cues used in production

and their role in processing is complicated. Specifically, Cangemi & Grice (2016) found that not only speakers use different combinations of cues (including those related to F0 and duration), but that there was not one optimal speaker with an optimal combination of cues; rather, listeners were sensitive to different speakers' combinations of cues. Thus, one speaker may be optimal for one listener in encoding an information structural contrast, but difficult to decode for another; that same listener, by contrast, would be able to optimally decode a different speakers' cue combination.

A related question has to do with whether during processing listeners rely on the cues they themselves employ in production. Though research suggests that production and perception are connected (e.g., Krivokapić & Byrd, 2012, Katsika et al., 2023), the link is not absolute, in that listeners do not preferentially attend to the cues they themselves use (Kim, 2020). This is not necessarily surprising: if listeners relied mostly on the cues they themselves used when processing the speech of others, communication could be hindered. Moreover, accommodation would be unlikely to occur if listeners were not able to pay attention to cues they do not habitually use themselves.

Current research also endeavours to find the sources of individual variability in processing, with results suggesting that individual differences may be due to characteristics of the speakers, such as autistic-like traits, empathy, musicality, and even gender. This had been previously shown for segmental aspects of the speech signal (see Yu and Zellou, 2019, for an overview). For example, several studies have linked the ability of speakers to imitate others and pay attention to phonetic detail to the extent in which they exhibit autistic-like traits, as measured by the Autism Spectrum Quotient (Baron-Cohen et al., 2001) or its subscales (e.g., Stewart & Ota, 2008; Yu, 2010; Yu, Abrego-Collier & Sonderegger, 2013). Similar results have been more recently reported for prosody as well. Bishop et al. (2020) found that participants with higher scores in the AQ communication subscale were less likely to mark unaccented words and words with prenuclear accents as prominent relative to those with lower scores. These authors also found that men were less likely than women to mark words as prominent. Similarly, Orrico et al. (2025) found that those of their RPT study participants who scored high in musicality (as measured by the Mini-PROMS test; Zentner & Strauss, 2017) were more likely to mark L+H* accents as prominent relative to participants with low scores, while those with high AQ scores were more likely to mark H* accents as prominent (contra Bishop et al., 2020, who report lower sensitivity to accent for high AQ participants).⁶ Empathy Quotient (Baron-Cohen and Wheelwright, 2004) was also employed to identify group perception differences for contrastive pitch accent identification in French: Estève-Gibert et al. (2020) conducted an eye tracking study in which the high empathy group employed intonation early in utterance processing to identify a homonymous but contrastive lexical item. Orrico et al. (2025) found that participants

⁶ It is important to note that reporting that participants have relatively high Autism Quotient scores does not mean that they are autistic. A study with participants diagnosed with Autism Spectrum Disorders (Grice et al., 2016) found that decisions as to whether a word sounded prominent (in the sense of “new”) was primarily based on invariant aspects of the lexical item (word frequency, animacy) rather than on accent type, reflecting a general tendency to pay less attention to intonation when decoding pragmatic aspects of the message (Grice et al., 2023).

who scored high on empathy were more likely to pay attention to pragmatic factors when assessing prominence. The results of Orrico et al. (2025) echo those of Baumann & Winter (2018), whose German participants could be classed into two groups, those who assessed prominence by responding primarily to pitch-related cues and those who largely responded to signal extrinsic aspects of the stimuli.

7. Updating the AM understanding of intonation

The research we have reviewed here shows that, by and large, the AM understanding of intonational phonology stands: tunes consist of independent elements that phonologically associate with phrasal boundaries and (for languages that rely on them) metrical heads. Phonetically, these associations are reflected in the synchronisation of tune and text. The phonological elements encode primarily pragmatic meaning, but also reflect connections to syntax (Hirschberg, 2006).

At the same time, there are several aspects of AM that need revising in light of recent research. With respect to phonology, one such aspect is that potentially the categories posited by many AM models are too fine-grained. As mentioned in section 6.3.1, the imitation studies of Cole et al. (2023) and Steffman and Cole (2024) suggest that most speakers may not distinguish as many tonal categories – either accentual or phrasal – as posited by Pierrehumbert (1980) and subsequent analysis of Mainstream American English (cf. Roy et al., 2017, discussed in 6.1.1). Unnecessary granularity may apply to the analyses of other languages too, leading to difficulties in settling on a consensus system. Thus, aiming for broader abstract categories and resisting the temptation to discretise small differences observed in highly controlled spoken data may be a better approach, as discussed in section 6.1.2 (cf. Arvaniti, 2016; 2019; Arvaniti et al., 2024). Highly controlled spoken data not only amplify small differences in the absence of contextual variability, but may also fail to capture what speakers actually do when communicating, i.e., when unscripted speech is considered (Grice et al., 1997; Cangemi et al., 2023; Arvaniti et al., to appear).

In order to better address the difficulties encountered in determining a language's tonal categories, we also need to re-evaluate our methods and heuristics. Since the 1990s much research focused on tonal alignment and a strong version of the Segmental Anchoring Hypothesis. It is now clear that this is problematic, especially if used without recourse to additional evidence. First, phonetic realisation is not invariant but subject to variation due to different sources, including the tonal environment, the segmental structure of the text, phonetic parameters such as speaking rate, sociolinguistic and paralinguistic influences, and individual variation. Second, recent research suggests that, like other phonological categories, intonation categories are realised by means of multiple cues. Besides aspects of F₀, these include changes in the periodic energy, duration and voice quality of segments, which may enhance the effect of F₀ (Albert et al., 2018) or may even be in a cue-trading relationship with it (Arvaniti et al., 2024), though more research is needed to better document and understand these effects. Finally, a number of studies have shown that not all putative targets behave in a similar fashion, especially in situations of tonal crowding (Arvaniti et al., 2017; Torreira & Grice,

2018), and its opposite, metrical enhancement (Steffman & Cole, 2024). As these authors have argued, such differences can be used as criteria for determining an intonation system's phonological structure.

In section 6.2, we presented a few promising methods for examining data, many of which are data-driven and thus can help avoid issues with annotation agreement, on the one hand, and the possibility of circularity and confirmation bias in annotation practices, on the other. We stress that each of these methods – TCoG, ProPer, GA(M)Ms, or FPCA – can be particularly helpful in answering specific questions about intonation; e.g., while TCoG and ProPer link production with perceptual properties that can help researchers determine which of their posited categories are perceptually relevant in the system under analysis, FPCA can help determine the essential dimensions of variation and separate systematic (linguistically-driven) variation from random variability or noise. In short, there is not one correct way of conducting intonation analysis. Indeed, as we have argued, using predominantly one methodology can be limiting; triangulation is likely to lead to more robust findings and has already led to greater insights regarding the phonetics of intonation.

We think that the methods we have reviewed will remain useful regardless of whether the field eventually adopts a dynamical systems approach such as those recently proposed by Roessig et al. (2019) and Iskarous et al. (2024). For example, while Iskarous et al. (2024) argue that intonation systems emerge directly from the selection of their model's parameters and their value ranges, it is not possible to dispense with other methodologies altogether: it is still the case that models such as that of Iskarous et al. will have to be compared to empirical data for validation (as Iskarous et al., 2024, do). For such comparisons to be meaningful and apt, using suitable techniques for the compilation and analysis of spoken corpora is also important. It goes without saying that perceptual validation is a critical additional step.

Finally, we note that one of the reasons why progress in intonation research lags behind our knowledge about segmental aspects of speech is intonation's presumed exceptionalism. Intonation has often been seen as peripheral and as not exhibiting the systematicity we otherwise see in the structure of speech (see Arvaniti et al., 2024, for a discussion). Yet, many aspects of intonation that are considered problematic (e.g., individual variability, context-dependent variation, cue redundancy, category overlap, hypo-articulation during tonal crowding) have segmental equivalents which are readily accepted as non-problematic. Many of the studies reviewed here show that intonation is no exception and thus that it should be treated using similar criteria to those adopted for the study of segmental aspects of speech.

8. Remaining challenges and future developments

We hope that the above discussion has demonstrated that intonation is essential for understanding speech and oral communication, but it is also a challenging topic of research. Great strides have been made in the past quarter of a century, but a great deal remains to be discovered. Here we highlight four topics that, in our view, require more attention. We stress they are by no means the only ones, and perhaps reflect our own

interests coupled with our focus on phonetics and phonology, rather than neurophysiological aspects of intonation processing, neurodevelopmental aspects, intonation acquisition, or intonation in L2/L3 learning and teaching, all of which also deserve further study.

Intonation meaning remains a challenging area. Whilst segmental phonological categories can be determined on the basis of simple meaning-based tests, such as using minimal pairs, this is not possible in intonation, the meaning of which is largely semantic (topic, focus) and pragmatic (information status, illocutionary force) and most likely probabilistically computed (Calhoun, 2010, Kurumada & Roettger, 2022). Moreover, in intonation, the same channel is used to convey multiple meanings. Although formal models of intonation meaning exist, as briefly discussed in section 3.4, few are comprehensive and virtually none is available for languages other than English, let alone tested with spontaneous data, i.e., when intonation is actually being used for communication (but see Arvaniti et al., to appear, for a recent attempt). For a breakthrough to happen, a change in approach is necessary. Existing research on intonational meaning tends to follow one of two approaches. It either focuses on intonation form while relying on rather informal descriptions of meaning (characterising tunes as *condescending*, *friendly* and the like), or it treats intonation as quasi synonymous to F0, i.e., as a phonetic parameter in service of specific semantic or pragmatic functions, such as focus. The latter type of research often does not rely on models of intonation, such as AM, but focuses instead on measuring F0 and other phonetic parameters (e.g., Breen et al., 2010). Finally, the meaning of tunes is often investigated out of context and without consideration of communicative situations that are likely to lead to naturally produced tunes and realistic processing of their pragmatic implications. To avoid these issues, we contend that greater interdisciplinary collaboration is needed, so that research follows rigorous standards in all aspects of each study. This should be coupled with the development of realistic research paradigms adapted to the features of intonation (such as its probabilistic interpretation), as in the promising approach of Kurumada & Roettger (2022), and the study of spontaneous speech, which, thanks to the methodologies discussed in section 6, is now easier to handle.

A second topic on which relatively little progress has been made so far is intonational typology. The reviews of Zerbian (2010) and Jun (2014b) are now 10 or more years old but have not yet been superseded, though studies that are smaller in scope are available, providing typological accounts of dialectal variation within a language (e.g., Bruce, 2005, on Swedish, Igarashi, 2014, on Japanese), geographically based accounts (e.g., Féry, 2010, on the languages of India), and studies of specific phenomena (e.g., Rialland, 2007 and 2009, on the intonation of questions in tonal languages of Africa). The problem of developing a general and generally accepted typology is compounded by the fact that, as noted, neither determining inventories nor understanding the phonetics of intonation are straightforward, and the same applies to the use and frequency of tunes. One distinction that is now widely accepted is, as noted in the introduction and section 3.1 as well, the distinction between languages in which melodic elements include both pitch accents and edge tones and those that use only the

latter. This issue is addressed in the typologies of Jun (2005c and 2014b), who couches the distinction in terms of head vs. edge-prominence, and Féry (2017), who makes a similar distinction in terms of intonational vs. phrasal languages. It is clear that greater insight will be gained as more languages and more varied systems are being investigated, including more in-depth investigation of the role of intonation in languages that use F0 for lexical purposes (tone and lexical pitch accent distinctions), as both have been known for some time to interact with intonation (e.g., Bruce, 1977, on Swedish; Pierrehumbert & Beckman, 1988, on Japanese; Peng et al., 2005, on Mandarin; Wong et al., 2005, on Cantonese; Joo & D'Imperio, 2023; see also Downing & Rialland, 2017, for a collection of papers on tone and intonation).

As mentioned throughout this paper, a great deal of intonation research has taken place in the laboratory. Consequently, it is not always clear how posited categories are realised in spontaneous speech or how they are processed during conversation, but existing studies suggest that caution is needed when we assert the generalizability of laboratory data. Apart from the fact that the distribution of intonational categories in read speech differs considerably from that in spontaneous speech (Grice et al., 1997), the boundaries between categories have been shown to be fuzzier in the latter, especially when the speech investigated has not been recorded for the purposes of phonetic research. This is evident in Cangemi et al. (2023), who investigated questions in a corpus of British English speech and found that the majority of questions had no clear intonational marking at all. Similarly, Gryllia et al. (2023), investigated H* and L+H* accents in Greek unscripted speech, and Arvaniti et al. (to appear) examined Greek wh-question tunes, as used in spontaneous data; both studies found that the clear distinctions evident in laboratory speech (e.g., Arvaniti et al., 2024; Gryllia et al., 2018; Baltazani et al., 2020) were more variable and fuzzier in spontaneous speech. For wh-questions in particular, Arvaniti et al. (to appear) found that tunes often served several pragmatic purposes at once. Although such differences should not come as a surprise, they are a challenge and an aspect of intonation that requires attention, as it can help us understand both intonational phonetics and phonology, and the role of intonation in communication.

A final aspect of intonation that we think deserves more attention is sociolinguistic variation. For some, intonation differences across varieties are negligible; e.g., Ladd (2022: 249) argues that intonation shows a “high degree of uniformity [...] across most varieties of ‘English’” and thus that differences across varieties need not be of great concern to researchers. However, comprehensive descriptions and studies across varieties of a given language indicate that differences can be substantial (see, e.g., Frota & Prieto, 2014, for comprehensive descriptions of dialectal variation in Portuguese, Catalan, Spanish, and Friulian; Gili Fivela et al., 2015, for comparisons among Italian varieties; Grabe et al., 2000, on dialectal differences in British English with respect to tonal crowding; Holliday, 2019, for a description of African American English patterns; Burdin, Holliday & Reed, 2022, on American English varieties; Grice et al., 2020, for an overview of English intonation). Thus, if ignored, such differences may add substantial noise in the data, while their phonological and pragmatic significance may be obscured. Yet, studies of

dialectal variation are few and far in between, while it is still unclear to what extent the differences affect communication, or what role they play in language change (but see Baltazani et al., 2022, and Baltazani et al., 2023, on contact between Greek and Venetian and Greek and Turkish respectively, and O'Rourke and Baltazani, 2023, for an overview of intonation in contact).

9. Conclusion

In conclusion, we have argued that intonation plays a central role in speech, language, and communication, as it is part of a language's phonology and interfaces with syntax and pragmatics. Thus, it should have a central role in the study of the above phenomena. Intonation is, however, challenging to study. Here, we presented the essential tenets of and empirical evidence for AM and focused on newer methodologies developed in the past 15 years or so in response to challenges brought about by emerging research findings. Though we have endeavoured to present a comprehensive and critical overview of newer methodologies, it would be impossible to do justice to them all. We remain confident, however, that the methods presented here hold promise, as they have already led to new insights in how intonation is structured and understood. Nevertheless, a great deal still remains to be done. One way forward, in our view, is to provide better training on intonation theory and methods, as very often problems stem from attempts to tackle intonation as simply a signal that is straightforwardly presented in acoustic analysis applications, rather than as a complex grammatical phenomenon that interfaces and interacts with other elements of grammar and speech. To achieve this goal, we advocate for more training in the form of dedicated summer schools, and workshops and tutorials as satellite events to major conferences. We hope that such initiatives will proliferate, leading to better research training as well as fruitful methodological innovations and theoretical insights.

CRedit authorship contribution statement

Amalia Arvaniti: Writing – review & editing, Writing – original draft, Visualization, Project administration, Investigation, Funding acquisition, Conceptualization. **Martine Grice:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Conceptualization. **Mariapaola D'Imperio:** Writing – original draft, Conceptualization.

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